

Deep learning high resolution burned area mapping by transfer learning from Landsat-8 to PlanetScope

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ABSTRACT

High spatial resolution commercial satellite data provide new opportunities for terrestrial monitoring. The recent availability of near-daily 3 m observations provided by the PlanetScope constellation enables mapping of small and spatially fragmented burns that are not detected at coarser spatial resolution. This study demonstrates, for the first time, the potential for automated PlanetScope 3 m burned area mapping. The PlanetScope sensors have no onboard calibration or short-wave infrared bands, and have variable overpass times, making them challenging to use for large area, automated, burned area mapping. To help overcome these issues, a U-Net deep learning algorithm was developed to classify burned areas from two-date PlanetScope 3 m image pairs acquired at the same location. The deep learning approach, unlike conventional burned area mapping algorithms, is applied to image spatial subsets and not to single pixels and so incorporates spatial as well as spectral information. Deep learning requires large amounts of training data. Consequently, transfer learning was undertaken using pre-existing Landsat-8 derived burned area reference data to train the U-Net that was then refined with a smaller set of PlanetScope training data. Results across Africa considering 659 PlanetScope radiometrically normalized image pairs sensed one day apart in 2019 are presented. The U-Net was first trained with different numbers of randomly selected 256×256 30 m pixel patches extracted from 92 pre-existing Landsat-8 burned area reference data sets defined for 2014 and 2015. The U-Net trained with 300,000 Landsat patches provided about 13% 30 m burn omission and commission errors with respect to 65,000 independent 30 m evaluation patches. The U-Net was then refined by training on 5,000 256×256 3 m patches extracted from independently interpreted PlanetScope burned area reference data. Qualitatively, the refined U-Net was able to more precisely delineate 3 m burn boundaries, including the interiors of unburned areas, and better classify “faint” burned areas indicative of low combustion completeness and/or sparse burns. The refined U-Net 3 m classification accuracy was assessed with respect to 20 independently interpreted PlanetScope burned area reference data sets, composed of 339.4 million 3 m pixels, with low 12.29% commission and 12.09% omission errors. The dependency of the U-Net classification accuracy on the burned area proportion within 3 m pixel 256×256 patches was also examined, and patches <6.5% burned were less accurately classified. A regression analysis between the proportion of 30 m grid cells classified as burned against the proportion labelled as burned in the 3 m reference maps showed high agreement ($r^2 = 0.91$, slope = 0.93, intercept <0.001), indicating that the commission and omission errors largely compensate at 30 m resolution.

1. Introduction

Wildland fires have substantial effects on terrestrial ecosystems and greenhouse gas emissions with a recent apparent surge of destructive fires causing social disruption and economic costs (Balch et al., 2017;

Bowman et al., 2020; Ward et al., 2020). Satellite data have been used for several decades to monitor fire, by detecting the locations of actively burning fires (Wooster et al., 2021) and by mapping the spatial extent of the area affected by fire, usually referred to as the “burned area” (Chuvieco et al., 2019). Many burned area products and algorithms have

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been developed for a variety of sensors, primarily using optical wavelength time series to detect the spectral changes induced by fire using per-pixel rule-based or supervised classification approaches. There is no consensus burned area mapping algorithm, and algorithms are sensor specific. In the initial remote sensing era, satellite based burned area mapping used one or a small number of medium spatial resolution Landsat images (Hall et al., 1980; Chuvieco and Congalton, 1988; López García and Caselles, 1991) or coarse spatial resolution Advanced Very High-Resolution Radiometer (AVHRR) data that are suboptimal for burned area mapping (Giglio and Roy, 2020). The most widely used systematically generated fire products are derived from Moderate Resolution Imaging Spectroradiometer (MODIS) data acquired by NASA's Terra and Aqua satellites that have dedicated fire monitoring capabilities (Justice et al., 2002; Giglio et al., 2018). However, the MODIS 500 m burned area product and other coarse spatial resolution burned area products do not detect small and spatially fragmented burns (Laris, 2005; Roy et al., 2019). It is well established that fire effects are non-permanent, for example, a recent global analysis with MODIS data found that the median burned area persistence time was only 29 days (Melchiorre and Boschetti, 2018). This and cloud and optically thick smoke obscuration mean that near-daily satellite observations, such as provided by MODIS or by combination of data from the Landsat and Sentinel-2 sensors, are required to reliably map on a systematic basis the area burned and the date of burning (Giglio et al., 2018; Roy et al., 2019). Since the advent of free and open access Landsat and Sentinel-2 data (Zhu et al., 2019), 20–30 m burned area products have been derived using time series algorithms applied over large areas including algorithms using data from Landsat combined with MODIS active fire detections (Boschetti et al., 2015), Sentinel-2 combined with MODIS active fire detections (Roteta et al., 2019), only Landsat (Hawbaker et al., 2017), and both Landsat and Sentinel-2 (Roy et al., 2019). The availability of high spatial resolution (< 10 m) satellite data provided by the commercial sector provides new opportunities for detailed burned area mapping and for assessment of post-fire damage and burn severity (Arnett et al., 2015; Dragozi et al., 2016; Warner et al., 2017; Vanderhoof et al., 2018). However, satellite data with high spatial and high temporal resolution are only recently becoming available. Notably, Planet Labs Inc. has deployed a constellation of PlanetScope CubeSats that provide multispectral data at 3–4 m spatial resolution (Planet Team, 2021) with a 30 h global median average revisit interval (Roy et al., 2021).

This study demonstrates, for the first time, the potential for automated PlanetScope 3 m burned area mapping. Recently we illustrated qualitatively the potential of PlanetScope imagery for mapping small and spatially fragmented burns (Roy et al., 2019) and a number of other studies have demonstrated the utility of PlanetScope data for wildfire damage assessment (Michael et al., 2018; Chung et al., 2020). The high spatial and temporal resolution observations provided by the PlanetScope constellation may enable mapping of small and spatially fragmented burns, not detected at moderate or coarse spatial resolution, and may provide data needed to assess the accuracy of coarser resolution burned area products (Roy et al., 2019). However, there are several challenges: (i) the PlanetScope sensors acquire imagery in only the visible and near infrared (NIR) (and also red-edge in the latest sensor generation), and not in the short-wave infrared (SWIR) that is useful for burned area mapping (Pereira et al., 1999; Roy et al., 2005a; Bastarrika et al., 2011; Huang et al., 2016), (ii) imagery are acquired with quite different overpass times and so solar geometry and bi-directional reflectance effects (Huang and Roy, 2021; Roy et al., 2021) that may impact the ability to map burns reliably in single images and particularly when multi-temporal images are used (Trigg et al., 2005; Roy et al., 2019), (iii) radiometric inconsistencies may occur among images due to relatively low sensor calibration accuracy and spectral response function differences among the PlanetScope sensor generations (Houborg and McCabe, 2018; Huang and Roy, 2021). To help overcome these issues, a deep learning approach is described that classifies burned areas detected

in spatially coincident two-date PlanetScope image pairs that have been relatively radiometrically normalized. The deep learning approach, unlike conventional burned area mapping algorithms, is applied to image spatial subsets and not to single pixels and so incorporates spatial as well as spectral information.

Deep learning is increasingly used for satellite image classification (Zhu et al., 2017; Ma et al., 2019). Convolutional neural networks (CNN) were developed to explore spatial relationships to classify natural images into a single class (LeCun et al., 2015). The pixels in an image may be classified independently by application of a trained CNN to image spatial sub-sets, referred to as patches, to classify the central patch pixel. This process can be applied to patches translated across the image but is computationally expensive and can result in smoothed classified feature boundaries (Zhang et al., 2018; Martins et al., 2020). The U-Net architecture is a form of fully convolutional network that classifies every patch pixel and is less affected by these issues (Ronneberger et al., 2015). The U-Net is the deep learning architecture used in this study for these reasons and because it is widely used for satellite image classification (Xu et al., 2018; Stoian et al., 2019; Brandt et al., 2020). The U-Net has not been used to map burned areas in PlanetScope data, but has been used recently for burned area mapping with lower spatial resolution satellite data. Recently, Knopp et al. (2020) used U-Net to classify 10 m burned areas in 110×110 km post-fire Sentinel-2 images located in Italy, United Kingdom, and Spain, with a training dataset composed of ~ 2600 256×256 10 m Sentinel-2 patches and visually interpreted class labels. Independent reference patches were used to validate the results with a high burn class F1-score > 0.86. The F1-score, also known as the Dice coefficient, is the harmonic mean of the user's and producer's accuracies, sometimes referred to as the precision and recall, respectively (Congalton and Green, 2019). Pinto et al. (2020) implemented a U-Net architecture with a Long Short-Term Memory layer to map burned areas using 7 months of near daily 750 m Visible Infrared Imaging Radiometer Suite (VIIRS) images resampled to 0.01° for sub-national regions in California, Portugal, Brazil, Mozambique, and Australia. The U-Net was trained with 2000 128×128 0.01° pixel VIIRS patches labelled using the 500 m NASA MODIS burned area product. The results were compared with a variety of burned area products with variable burn class F1 scores from 0.582 to 0.920. Recent research has used other deep learning approaches to classify burned areas using Sentinel-2 and synthetic aperture radar Sentinel-1C-band data. Sentinel-1 data can penetrate clouds and smoke (Torres et al., 2012) but many sensing and environmental factors affect the C-band backscatter (Beaudoin et al., 1990; Pulliainen et al., 1996) and, for example, the change in backscatter pre- and post-fire can be comparable to changes in moisture content and seasonal phenology in savannas (Mathieu et al., 2019). Belenguier-Plomer et al. (2021) applied two CNN-based architectures independently to Sentinel-1, Sentinel-2, and to both datasets together, degraded to a common 40 m resolution to map burned areas in 110×110 km tiles in different biomes. The results were validated with 30 m Landsat burned/unburned area maps, and provided 0.35 to 0.64 F1 values that varied with land cover and the input data type. Similarly, Zhang et al. (2021a) trained a CNN with native 20 m Sentinel-1 and 20 m Sentinel-2 data to map burned areas in 12 fire events in California. The results were validated with PlanetScope 3 m reference data generated by thresholding NDVI difference maps and they reported F1 values from 0.77 to 0.96.

Supervised classification requires significant amounts of training data that are time-consuming and expensive to collect (Wulder et al., 2018) and in particular this is a bottleneck for deep learning (Najafabadi et al., 2015). One strategy to mitigate this issue is to use transfer learning, i.e., to train a classifier using pre-existing training data and then apply it to classify a different data set (Tuia et al., 2016; Nogueira et al., 2017; Zou and Zhong, 2018). The “black box” nature of deep learning algorithms precludes understanding of their internal decisions, but their generalization ability is known to enable repurposing for other classification tasks and data types (Oquab et al., 2014; Hu et al., 2015;

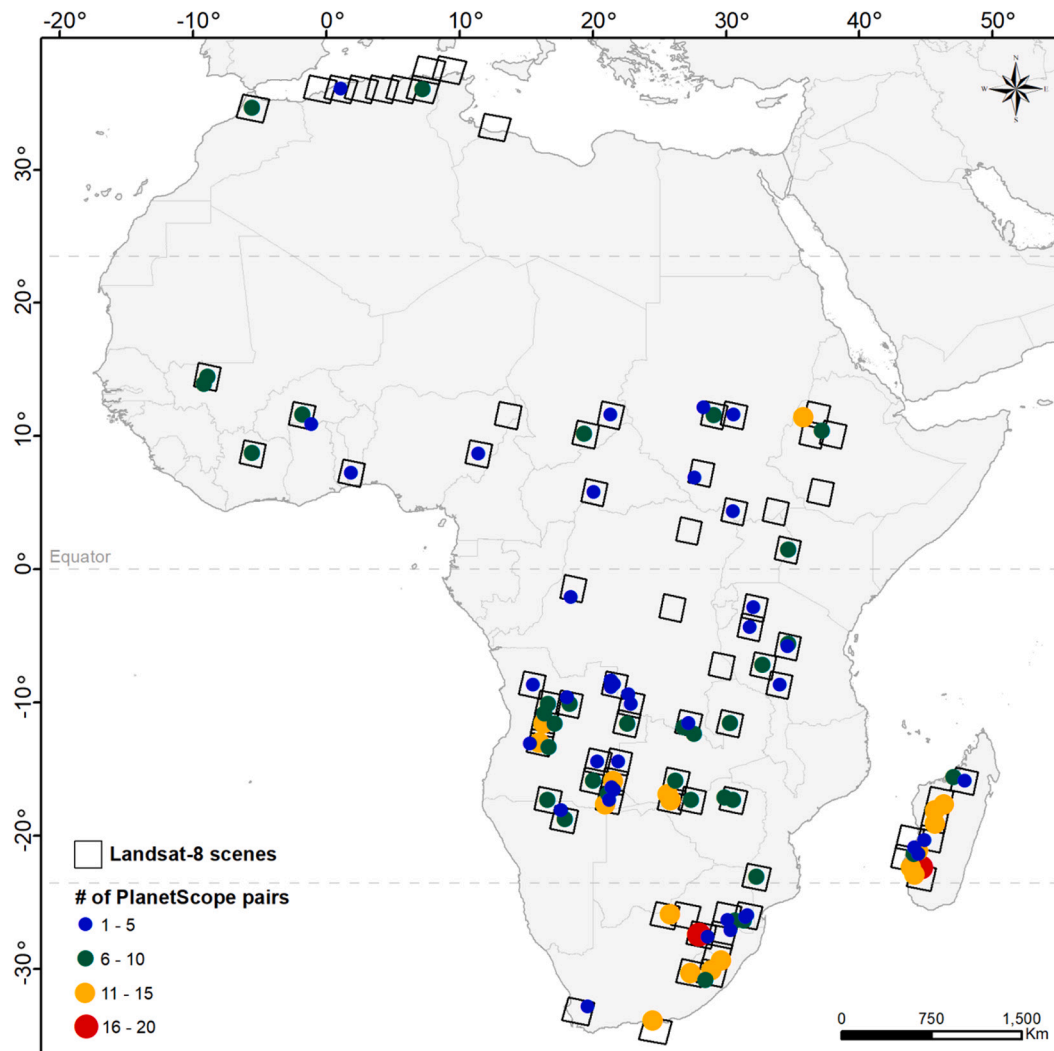


Fig. 1. Locations of the 92 Landsat 30 m burned area reference data derived from Landsat image pairs sensed 16 days apart at 78 unique Landsat path/rows (black squares) and of the 659 PlanetScope 3 m image pairs sensed one day apart (circles, colored to denote the number of pairs).

Marmanis et al., 2015). Usually, supervised classifiers become less reliable when used to classify satellite imagery acquired in space/time further from the conditions that the training data were collected under, or when applied to imagery that have different sensor characteristics or pre-processing. Despite this, studies have demonstrated successful transfer learning among different satellite sensors with broadly similar spectral characteristics (Wurm et al., 2019; Mateo-García et al., 2020). In this paper, a transfer learning approach is implemented by using a pre-existing 30 m Landsat-8 burned area data set to train a U-Net that is then refined with interpreted 3 m PlanetScope images. The research is demonstrated using 659 PlanetScope two-date image pairs distributed across Africa, which is the continent with the most burning (Giglio et al., 2018) and where we have experience developing and validating satellite burned area products (Roy and Boschetti, 2009; Roy et al., 2005a, 2005b, 2019). The refined U-Net is used to classify two-date PlanetScope image pairs, mapping the 3 m pixels that burned between the image acquisition dates. The results are validated with 20 visually interpreted burned area maps derived from 3 m PlanetScope image pairs acquired one day apart across Africa.

2. Data

2.1. Pre-existing Landsat-8 burned area reference data

The Africa portion (black squares, Fig. 1) of the Landsat-8 burned area reference data set generated to assess the accuracy of the Collection 6 NASA MODIS 500 m burned area product (Boschetti et al., 2019) was used to train the U-Net. At each location, the data set is composed of a visually interpreted 30 m burned area map and the two Landsat-8 Operational Land Imager (OLI) atmospherically corrected images sensed 16 days apart that were used to generate it. The Landsat-8 image pairs were selected between March 1st 2014 to March 19th 2015 following a probability-based sampling protocol using a stratified random sampling in both time and space (Boschetti et al., 2016). The Landsat-8 image pairs were interpreted visually following the Committee on Earth Observation Satellites (CEOS) protocol (Boschetti et al., 2009) to label 30 m pixels as either burned (i.e., locations burned between the two image acquisition dates), unburned (i.e., locations that did not burn between the two image acquisition dates), or unmapped (i.e., locations that were unobserved due to clouds or shadows in one or both images, or that could not be interpreted unambiguously). A total of 92 30 m burned area reference maps, and their associated Landsat-8 surface reflectance image pairs, located at 78 unique Landsat path/row locations were selected. All of them were selected to contain at

Table 1

The Landsat-8 30 m OLI and PlanetScope-0 and PlanetScope-1 3 m bands (and bandwidths, units in nm) used in this study.

	Green	Red	NIR
Landsat-8 OLI	530–590 (60)	640–670 (30)	850–880 (30)
PlanetScope-0	500–590 (90)	590–670 (80)	780–860 (80)
PlanetScope-1	547–585 (38)	650–682 (32)	846–888 (42)

least one burned 30 m pixel. They occur in each of the five African biomes (tropical savanna, temperate savanna, deserts and xeric shrublands, tropical forest, Mediterranean) defined by Olson et al. (2001). This pre-existing Landsat-8 burned area reference data set has several benefits for this study: (i) they were derived with high accuracy needed for training purposes, (ii) the sampling methodology used to select the data ensures a range of burned and unburned conditions, and (iii) the Landsat-8 OLI has high radiometric and geometric quality (Roy et al., 2014) and similar visible and NIR bands as the PlanetScope sensors (Table 1) which is helpful for transfer learning from Landsat to PlanetScope.

2.2. PlanetScope data

The first two sensor generations, PlanetScope-0 (Dove-Classic) and PlanetScope-1 (Dove-R), that acquire 12-bit blue, green, red, NIR radiance, were used as only a minority of the third generation (SuperDove) images were available when this research was initiated. Atmospherically corrected 3 m orthorectified images (Level 3B), each typically covering 25 km × 11.5 km (PlanetScope-0) or 25 km × 23 km (PlanetScope-1), were used. The data are atmospherically corrected with the 6SV radiative transfer model, the same used to correct the Landsat-8 OLI data (Vermote et al., 2016), with MODIS derived aerosol information (Planet Team, 2021). The PlanetScope 3 m Usable Data Mask (UDM2), that labels each pixel as clear, cloud, cloud shadow, haze, snow, missing, or suspect due to saturation or downlink errors (Planet Team, 2021), was used. When the UDM2 was not available, the Usable Data Mask (UDM) that has fewer quality flags (cloudy or suspect flags) was used. The shorter wavelength blue bands available on PlanetScope and Landsat-8 OLI were not used because atmospheric correction errors are significantly greater at blue wavelengths (Fraser and Kaufman, 1985; Ju et al., 2012) which makes them less reliable for burned area mapping (Roy et al., 2019).

A total of 659 PlanetScope 3 m image two-date pairs (dots, Fig. 1) acquired November 2018 to December 2019, i.e., in different years to the Landsat burned area reference data, were used. Only PlanetScope two-date image pairs that met the following criteria were used: (i) fall within or intersected the 78 unique Landsat path/row locations, (ii) have ≥ 25 km² common overlapping spatial area and ≥ 75% high-quality pixels (cloud- and shadow-free, no missing or suspect pixels) in both images (as labelled by the UDM2 or UDM), (iii) have < 3 m pixel misregistration assessed by visual comparison of the two images in each pair, (iv) be able to be relatively radiometrically normalized (Section 3.2.1), (v) be composed of images acquired one day apart and with solar zenith angle difference < 5°. Criterion (i) was to leverage the transfer learning with similar landscape characteristics in both Landsat and PlanetScope datasets, (ii) - (iv) were to ensure that a meaningful amount and good quality PlanetScope image pair data were classified, and (v) was to reduce unwanted bidirectional reflectance effects as the images are acquired with different overpass times and so different solar zenith angles (Huang and Roy, 2021).

3. Methodology

3.1. Overview

The methodology was implemented in the following manner: (i) a

relative radiometric normalization was applied to each PlanetScope 3 m image pair to minimize non-surface change differences associated with the variable image quality, (ii) two independent subsets of PlanetScope image pairs were interpreted into 3 m burned area reference data, (iii) two independent sets of 30 m pixel patches were extracted from the pre-existing Landsat-8 burned area reference data, one set was used to train the U-Net and the other to demonstrate the U-Net performance applied to Landsat data and to derive the optimal U-Net training size, (iv) the Landsat trained U-Net was refined with 3 m patches extracted from the first set of the PlanetScope burned area reference data to improve the applicability of the U-Net for the PlanetScope classification, (v) the refined U-Net was used to classify all 3 m PlanetScope image pairs, (vi) the classification accuracy was evaluated using the second set of PlanetScope burned area reference data, including an accuracy assessment of the scale discrepancy between the Landsat and PlanetScope imagery. These steps, and the U-Net architecture, are described below in detail.

3.2. Pre-processing and generation of PlanetScope and Landsat reference data

3.2.1. PlanetScope image pair relative radiometric normalization

The selected PlanetScope image pairs were sensed one day apart with < 5° solar zenith difference, and so differences between the images, other than surface changes due to any burning, were due to sensor bandwidth and spectral response differences (if the images were sensed by different PlanetScope sensor generations), calibration differences, and residual atmospheric correction errors (Huang and Roy, 2021). To minimize non-surface change differences, a relative radiometric normalization approach was applied independently to the red, green and NIR surface reflectance in each pair. The image in each pair with the smallest aerosol optical depth (AOD) (reported in the image metadata) was selected as the independent image and the other as the dependent image. For each band a Theil-Sen regression was used to estimate a linear model between the independent and the dependent image. The Theil-Sen estimator is less sensitive to outliers (Theil, 1950; Sen, 1968) and so the spectral normalization of no-change pixels becomes more reliable as pixels with abrupt surface change are often treated as “outliers” (Schott et al., 1988; Olthof et al., 2005). The regression was derived using corresponding pixel surface reflectance values sampled every 15 pixels east-west and north-south directions, discarding pixels that were flagged as cloud or cloud shadow in either image. The Pearson correlation coefficient (r) between the band pixel samples was derived, and if $r < 0.5$ for any of the three bands, or if the number of pixels considered was < 10,000, then the image pair was discarded. Otherwise, the Theil-Sen regression coefficients (slope and intercept terms) were applied to all the surface reflectance pixel values in the dependent image to normalize the image.

3.2.2. Generation of PlanetScope 3 m burned area reference data and extraction of 3 m burned area patch data that were used to refine the Landsat trained U-Net

PlanetScope 3 m burned area reference data were derived for 40 of the 659 radiometrically normalized image pairs. The 40 pairs were selected across all five Africa biomes and included 13 of the 17 International Geosphere-Biosphere Programme (IGBP) land cover classes defined in the Collection 6500 m MODIS land cover product for 2019 (Sulla-Menashe et al., 2019), with a diversity of burning conditions. Each of the 40 pairs were interpreted visually to label 3 m pixels as either burned (i.e., the pixel burned between the two image acquisition dates), unburned (i.e., the pixel did not burn between the two acquisitions), or unmapped (i.e., surface was unobserved due to clouds or shadows in one or both images, or could not be interpreted unambiguously). This was time consuming due to the complexity of the landscape and the subtle patterns of burning that can be captured in PlanetScope imagery (Roy et al., 2019). Therefore, to help speed up the interpretation, the second image acquisition was segmented into burned and

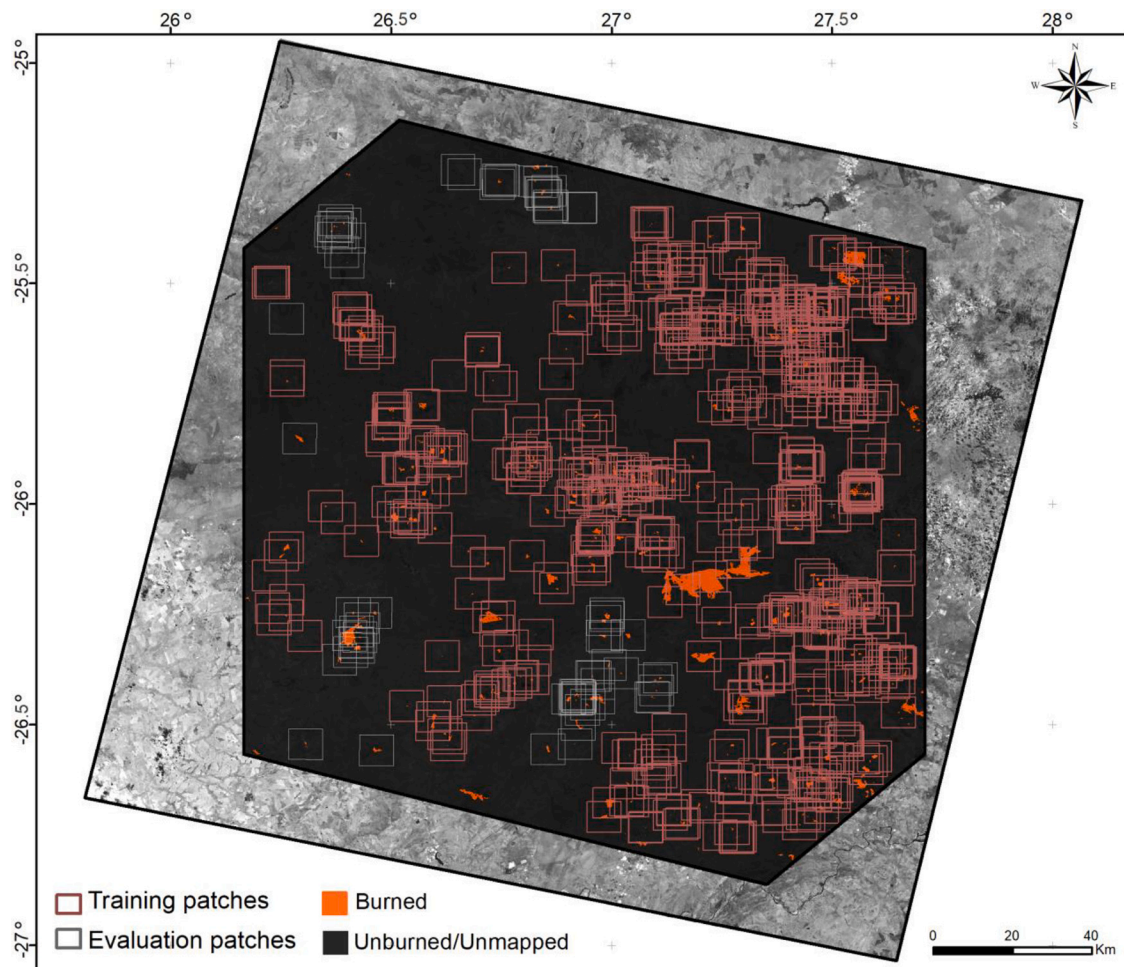


Fig. 2. Example 256×256 30 m Landsat training (654 red squares) and evaluation (73 gray squares) patches extracted from pre-existing Landsat-8 burned area reference data derived from Landsat-8 OLI images acquired October 6 and October 22 2014 over Gauteng Province, South Africa. The orange pixels show the 30 m pixels labelled in the Landsat-8 burned area reference data as burned (i.e., locations that burned between the two Landsat image acquisition dates), black shows pixels that were unburned (i.e., locations that did not burn between the two image acquisition dates) or were unmapped (i.e., locations that were unobserved due to clouds or shadows in one or both images, or that could not be interpreted unambiguously). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

unburned classes using the Mean-Shift segmentation algorithm (Comaniciu and Meer, 2002) applied to the green, red, and NIR bands. The image was over-segmented to provide many segments per burn, so that they could then be rapidly checked, edited, and interactively merged on-screen. This over-segmentation reflects standard practice (Pavlidis and Liow, 1990; Blaschke et al., 2014). Segments with new burn pixels occurring between the two image dates were selected and labelled as burned. A second interpreter was responsible for quality control through visual inspection, and if needed, further refinement of the burned area edges. The resulting 40 pairs of interpreted PlanetScope imagery were split into two halves - twenty pairs were used to validate the final PlanetScope burned/unburned classification (Section 3.4) and the other twenty were used to refine the U-Net model (Section 3.3.3) using 5000 extracted 256×256 3 m patches.

The U-Net trained with 256×256 30 m Landsat training patches was refined using 256×256 3 m patches extracted from the 20 PlanetScope burned area reference data. Note that 256×256 pixel patch dimensions are often used for U-Net remote sensing studies (Wagner et al., 2019; Brandt et al., 2020). A total of 5000 256×256 3 m patches were used to refine the Landsat trained U-Net. Specifically, from each of the 20 PlanetScope burned area reference data sets, 250 256×256 3 m patches composed of a predictor variable (a numeric code to denote burned or unburned) and six explanatory variables (green, red, and NIR surface

reflectance values from each of the two PlanetScope images sensed one day apart) were randomly extracted. In this random selection process, any patches containing 3 m pixels labelled as unmapped (i.e., surface was unobserved due to clouds or shadows in one or both PlanetScope images, or that could not be interpreted unambiguously) were rejected.

3.2.3. Generation of Landsat-8 30 m burned area patch data

A large number of 256×256 30 m pixel patches were extracted independently from each of the 92 pre-existing Landsat-8 burned area reference data sets (Section 2.1). The majority, termed Landsat training patches, were used to train the U-Net (Section 3.3.1) and a minority, termed Landsat evaluation patches, were used to derive the optimal Landsat-8 burned area patch training set size (Section 3.3.2). Each patch 30 m pixel was composed of six explanatory variables and a predictor variable. The six explanatory variables were the red, green and NIR surface reflectance values of the two Landsat-8 OLI images used to derive the Landsat burned area reference data, and the predictor variable was a binary code to denote if the pixel was burned or unburned as defined by the Landsat burned area reference map. Pixels labelled as unmapped in the Landsat burned area reference map (i.e., because they were unobserved due to clouds or shadows in one or both Landsat images) were labelled unburned. This is not an issue because the final U-Net classifications of the PlanetScope imagery are post-processed using

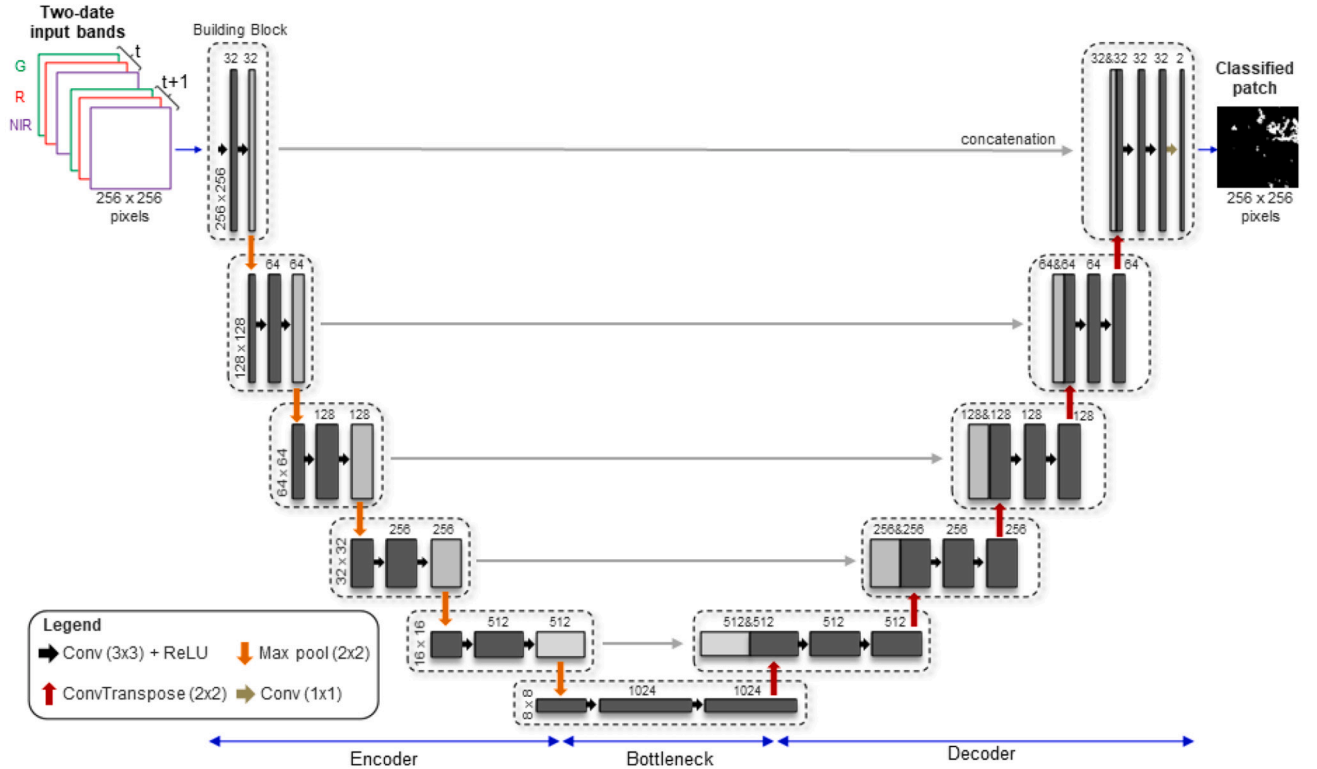


Fig. 3. The U-Net architecture used to classify 256×256 pixel patches (two dates of green, red, and NIR surface reflectance) into burned or unburned classes. Each building block (dashed black boxes) contains two convolutional layers that are used to generate feature maps (gray filled rectangles). The number of feature maps is specified above each gray filled rectangle and the feature map spatial dimensions are defined on the left side. The horizontal gray arrows show skip connections used to copy feature maps from the encoder (light gray rectangles) to their decoder block counterpart. The orange and red vertical arrows show the maximum pooling and transposed convolution operations, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the PlanetScope UDM2/UDM cloud and shadow mask (Section 3.3.4).

Fig. 2 shows a typical example of the training (red) and evaluation (gray) patches that were extracted. The patches were selected randomly from each Landsat-8 burned area reference data set but constrained to ensure that (i) each patch contained one or more 30 m burned pixels, (ii) the majority of the extracted patches were designated as training patches and the remainder as evaluation patches, and (iii) the training and evaluation patches did not overlap spatially to ensure that they were independent. The total number of extracted patches varied among the Landsat-8 burned area reference data sets (being dependent on the number and spatial arrangement of the reference data 30 m burned pixels). The patch selection process was as follows. First, the 30 m burned pixels were spatially clustered into n_{burned} burned areas, each composed of spatially connected burned pixels found by 8-neighbor adjacent pixel clustering. Then the geographic centroids of each of the n_{burned} burned areas were derived. Next, candidate extracted patches were defined by 256×256 30 m pixel boxes centered on each geographic centroid. All the boxes that overlapped spatially were identified and given a unique group membership number $\{1, 2, \dots, n_{\text{groups}} \leq n_{\text{burned}}\}$. Then the training and evaluation patches were selected randomly from different groups to ensure that the training and evaluation patches did not overlap spatially. This random selection was undertaken iteratively until 80–90% of all the boxes were selected as training patches and the remainder were assigned as evaluation patches. Sometimes, due to the spatial topology of the 30 m pixel burned areas, the 80–90% allocation was not possible. In this case, the largest group of boxes was divided into two or more independent groups by removing up to 50 boxes that overlapped. This allowed the generation of more non-overlapping groups. The above selection procedure was able to provide a large number of Landsat training and evaluation patches from

each Landsat-8 burned area reference data set. For example, in Fig. 2 a total of 654 training (red) and 73 evaluation (gray) patches were extracted.

3.3. U-Net deep learning

3.3.1. U-Net architecture and training with Landsat-8 burned area patch data

The U-Net uses the generative nature of a convolutional autoencoder to learn delicate boundaries with a multiscale architecture to provide different levels of data abstraction that are used to classify each patch pixel (Ronneberger et al., 2015). The architecture consists of three sections. i.e., an encoder, a bottleneck, and a decoder. The encoder (or contracting path, Fig. 3 left side) explores higher semantic information by reducing in steps the spatial resolution of the input patch data, while the decoder (or expanding path, Fig. 3 right side) recovers the spatial information.

The U-Net is composed of a sequence of building blocks (Fig. 3 dashed boxes) containing two convolutional layers. Each convolutional layer consists of kernels, defined by $i \times i \times r$ real valued matrices, termed kernel weights, and an associated single real valued bias, where $i \times i$ are the kernel spatial dimensions and r is the number of spectral bands in the input image patch for the first convolutional layer and for subsequent layers is equal to the number of feature maps in the previous layer. When the U-net is trained, the kernel weights and the bias values, collectively termed the network coefficients, are defined. In the U-Net classification process a series of square two-dimensional real valued matrices, termed feature maps, are generated (Fig. 3 gray filled rectangles). The core operation in the convolutional layer is a spatial convolution of the kernel matrix weights with the input patch data (in the first layer) or previous

feature map (in subsequent layers), followed by addition of the bias term to generate the feature map. Typically, every feature map value is non-linearly transformed and in this study the rectified linear unit (ReLU) function was used so that after the transformation any negative values were set to zero and positive values remained unchanged (Glorot et al., 2011).

The U-Net encoder section (Fig. 3 left side) consists of several contraction blocks. Each block takes an input that applies two convolution layers followed by a maximum pooling step. The maximum pooling calculates the maximum value of each 2×2 adjacent pixel region of the feature map to create a downsampled feature map. The number of kernels (and so feature maps) after each contraction block doubles so that the architecture can learn complex structures effectively. The U-Net bottleneck is the bottom-most layer and mediates between the encoder and the decoder. It uses two convolution layers without any pooling operation. The U-Net decoder section (Fig. 3 right side) consists of as many expansion blocks as there are encoder contraction blocks. Each expansion block takes an input that applies two convolution layers preceded by a transpose convolution step. The transpose convolution creates an upsampled feature map by inserting a column and a row of zeros between each row and column of the feature map to increase the feature map size and then applying a 2×2 convolution kernel with learnable weights. Skip connections (Fig. 3 horizontal gray arrows) are used to copy encoder feature maps (Fig. 3 light gray rectangles) to their decoder block counterparts to ensure that the features learned while contracting the image are used to reconstruct the image. In the last decoder block (top right, Fig. 3) there are an additional n convolutional layers (and so feature maps), where n is the number of desired classes, that are filtered to provide the final classification result.

Fig. 3 shows the U-Net structure used to classify each 256×256 pixel patch composed of two dates of green, red, and NIR surface reflectance into burned and unburned classes. Similar to most U-Net land cover classification studies, 3×3 kernels were used. Following common practice, 32 kernels for each of the two convolutional layers in the first encoder block (Wagner et al., 2019; Brandt et al., 2020) were implemented but with an additional block in the encoder and decoder (five instead of four) to explore deeper abstraction features. This provided blocks with 32, 64, 128, 256, 512, and 1024 kernels per convolutional layer that generated square feature maps with side dimensions of 256, 128, 64, 32, 16 and 8 pixels respectively. The last decoder block has an additional two layers that generate 256×256 burned and unburned feature maps.

The extracted Landsat training patches (Section 3.2.3) were used to train the U-Net to derive the network coefficients. A conventional gradient descent method was used to iteratively update the coefficients by minimizing a loss function. The network coefficients were initialized randomly, then with each iteration they were updated by adding the coefficient gradient values of the loss function perturbed by a small amount. The training was undertaken in two steps: “mini-batches” of training data were passed in the forward propagation through the network, and then, the estimated error between the predicted and training data class labels was used to update the coefficients during the backpropagation (Rumelhart et al., 1986; LeCun et al., 1990). This process can be described:

$$\mathcal{L}(\theta) = -\frac{1}{m} \sum_{i=1}^m (\hat{c}_i \times \log c_i + (1 - \hat{c}_i) \times \log(1 - c_i)) \quad (1)$$

where $\mathcal{L}$ is the binary loss function, θ are the network coefficients, m is the number of training patches in the mini-batch, c is a vector of the patch pixel training data class labels, and $\hat{c}$ is a corresponding vector of the U-Net predicted class confidences (real numbers from 0 to 1). The backpropagation algorithm computes the partial derivatives, i.e., $\partial \mathcal{L}(\theta) / \partial \theta$, of the loss function (LeCun et al., 2015), and the network coefficients are updated by gradient descent after each interaction (Bengio, 2012) as:

$$\theta^{(t+1)} = \theta^{(t)} - \eta \frac{\partial \mathcal{L}(\theta)}{\partial \theta} \quad (2)$$

where θ are the network coefficients, t is an iteration step equivalent to a single forward mini-batch of training data introduced to the network, and η is the learning rate defined by a small (< 1) positive real number. An epoch of iterations is completed when all the training patches are used, and several epochs are needed to update the network coefficients until a satisfactory classification performance is obtained.

The U-Net was implemented with the TensorFlow 2.2.0-Keras framework (Abadi et al., 2016) and configured with a conventional binary loss function and using Adam optimization of the gradient descent with default settings (learning rate $\eta = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$) (Kingma and Ba, 2015). Conventional batch normalization was applied to normalize the feature maps, using the mean and standard deviation of the feature maps derived from each mini-batch of training samples, to reduce overfitting and vanishing gradient issues (Ioffe and Szegedy, 2015). The training was undertaken over 75 epochs with a 128 mini-batch size. In practice, the number of epochs is evaluated during the training progress, too few epochs may result in poor classification accuracy and too many may lead to overfitting (Liu et al., 2008). In this study, using the optimal Landsat-8 burned area patch training data set (described in the next section), 75 epochs were used because the training overall accuracy converged and did not improve $> 0.05\%$ over the last 10 epochs.

3.3.2. Landsat U-Net burned/unburned classification and selection of optimal Landsat-8 burned area patch training set size

The U-Net classification defines for each patch pixel a burn confidence value:

$$\hat{c}(i, j) = 1 / \left(1 + e^{-a^{last}(i, j)} \right) \quad (3)$$

where $\hat{c}(i, j)$ is the burn confidence (a real number from 0 to 1), (i, j) is the patch pixel location, and $a^{last}(i, j)$ is the burn class feature map generated in the last decoder block (Fig. 3 top right). The burn confidence map was converted into a binary burned/unburned class value by application of a confidence threshold as:

$$b(i, j) = \begin{cases} 1, & \text{If } \hat{c}(i, j) \geq c_0 \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

where $b(i, j)$ is the binary classification (1 = burned, 0 = unburned), $\hat{c}(i, j)$ is the burn confidence defined as Eq. (3), and c_0 is the confidence threshold nominally set as 0.5.

Deep learning performance typically increases with more training data but at increasing computational cost (Garcia-Garcia et al., 2018; Zhang et al., 2021b). In general, if increasing amounts of training data provides reduced classification accuracy then the model is overfitting and if the classification accuracy continues to increase as training data are added then the model is still underfitting. Therefore, in this study, experiments were undertaken to assess the impact of using an increasing proportion (0.0125, 0.0625, 0.125, 0.1875, 0.25, 0.5, 0.75, 1.0) of the Landsat training patches to obtain an optimal set that provided a stable and accurate classification of the Landsat evaluation patch data. Specifically, the U-Net was trained independently with each training proportion selected randomly without replacement from all the Landsat training patches, and each trained U-Net was then used to classify all the Landsat evaluation patches. The six explanatory variables (three bands of Landsat surface reflectance, for two dates) in each Landsat evaluation patch were U-Net classified and the results compared with the evaluation patch predictor values (i.e., burned and unburned categories) to populate a burned/unburned confusion matrix. Note that the Landsat training and evaluation patches do not overlap (Fig. 2) and so the training patches used to train each U-Net never include the evaluation patches. Conventional overall classification accuracy and burn class

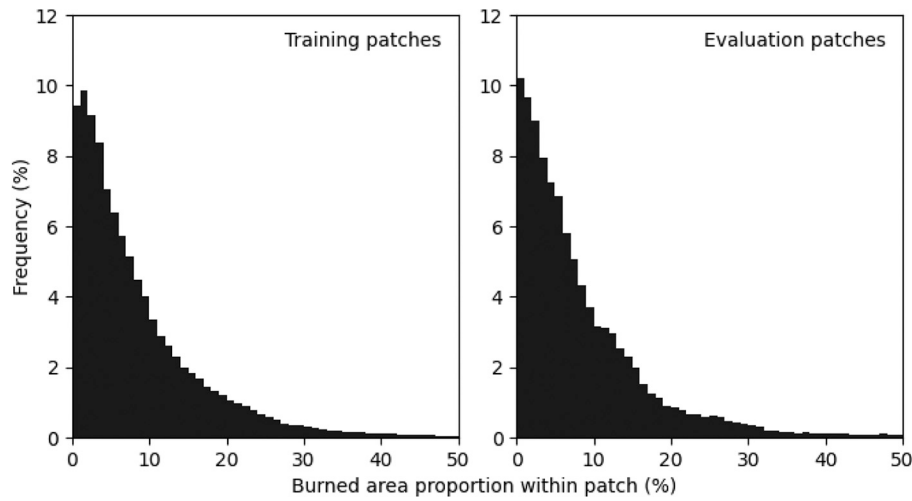


Fig. 4. Frequency distributions of the burned area proportion (expressed as a percentage of the 256×256 30 m pixel patch) of the 400,000 Landsat training and the 65,000 Landsat evaluation patches. Histogram bin widths equal 1%.

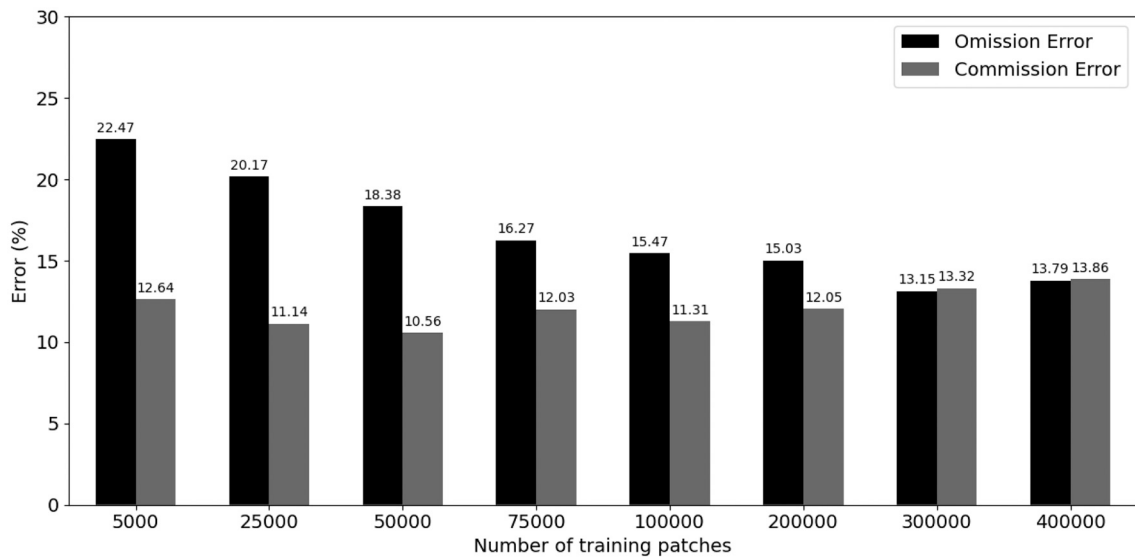


Fig. 5. Sensitivity analysis of the U-Net Landsat classification accuracy with respect to the number of Landsat training patches used to classify the 65,000 Landsat evaluation patches.

omission and commission error statistics were derived from the confusion matrices (Foody, 2002; Boschetti et al., 2019). The optimal set of training patches was selected as the one that provided high overall classification accuracy and a good balance between burn commission and omission errors. In these experiments c_0 was set as 0.5, i.e., the default threshold to convert the confidence to a binary class value.

3.3.3. U-Net refinement with PlanetScope burned area reference patch data

Initial classification results derived by applying the Landsat trained U-Net to the radiometrically normalized PlanetScope image pairs provided encouraging results (illustrated in Section 4). However, recent transfer learning research has demonstrated that augmenting, often termed fine-tuning, the training with a small set of training data derived from the imagery that will be classified (referred to as the target imagery) can improve the classification performance. There are a number of approaches including (i) replacement of the last convolutional layer by a new one with randomly initialized weights that are trained using target imagery training data (Agrawal et al., 2014; Campos et al., 2017), (ii) freezing some convolution layers and re-training the model on the

remaining layers using target imagery training data (Nogueira et al., 2017; Lima and Marfurt, 2020), or, (iii) undertaking additional training of the trained network layers using target image training data and a small learning rate (Tremblay et al., 2018; Cetinic et al., 2018; Wurm et al., 2019). In this study, approach (iii) was adopted because it enables tuning of the network coefficients with a relatively small target training data set and without overfitting (Maggiori et al., 2016; Nogueira et al., 2017). The U-Net trained with the optimal proportion of Landsat training patches was further trained with the 5000 PlanetScope 256×256 3 m patches extracted from the 20 PlanetScope burned area reference data sets developed for this purpose (Section 3.2.2). The U-Net was further trained with 25 epochs and a $1e^{-5}$ learning rate. The resulting U-Net is hereafter referred to as the *refined U-Net*.

3.3.4. PlanetScope burned/unburned classification using the refined U-Net

The refined U-Net was applied to classify each of the 659 radiometrically normalized PlanetScope 3 m image pairs. Each pair had a ≥ 25 km² common overlapping spatial area and so the refined U-Net was applied to 256×256 3 m pixel subsets translated systematically every

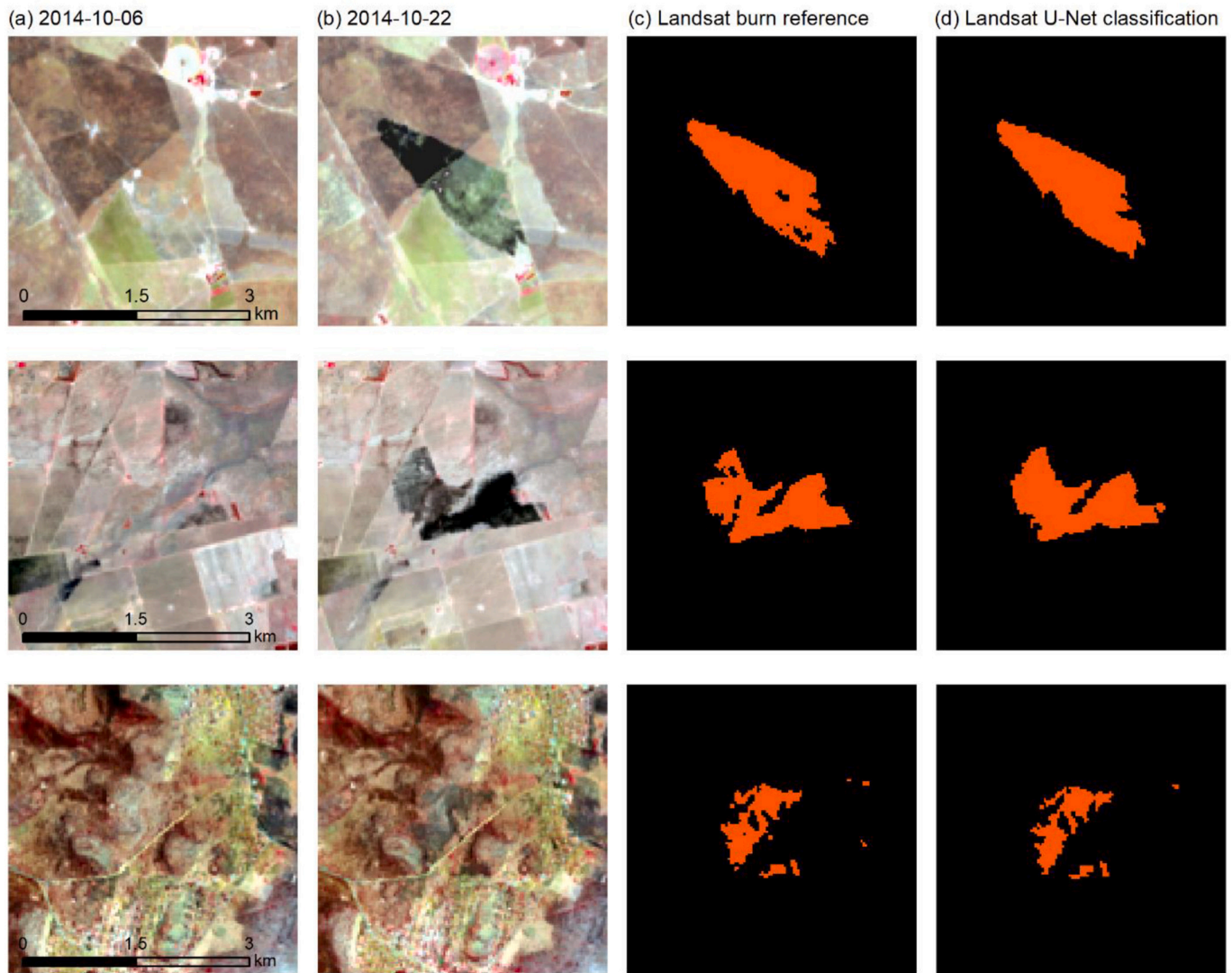


Fig. 6. Example Landsat-8 NIR-Red-Green surface reflectance 30 m images acquired 16 days apart on (a) October 6 2014, and (b) October 22 2014, (c) the corresponding Landsat evaluation patch data, (d) the U-Net classification of the two Landsat-8 images. Results for three 128×128 30 m pixel subsets selected within three Landsat evaluation patches located in the north, west, and south of Fig. 2. The U-Net was trained with 300,000 Landsat training patches. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

192 pixels across the common area so that the subsets overlapped by 25% (i.e., by 64 3 m pixels) in the x and y image axes. An overlap of this magnitude is often used because the U-Net classification performance may be reduced nearer the sub-set borders (Ronneberger et al., 2015; Derksen et al., 2019). The refined U-Net was applied to each subset pixel to derive a burn confidence value (Eq. (3)) that was thresholded (Eq. (4)) into a burned or unburned class. Classified pixels falling within 16 pixels of each subset border were discarded (Freudenberg et al., 2019) and overlapping areas among the different classified subsets were combined by voting (Li et al., 2019) to keep burn classified pixel values. Pixels labelled as cloud, cloud shadow, missing or suspect in the UDM2/UDM of either image in the pair were reclassified as “unmapped”. Thus, the PlanetScope classification had three classes: burned, unburned, and unmapped.

The final classification results were generated using an optimal confidence threshold (c_0). The threshold was determined by thresholding the burn confidence values generated by application of the refined U-Net to the 20 PlanetScope burned area reference data sets used to refine the U-Net (Section 3.3.3) with different c_0 values (from 0.1 to 0.9, in steps of 0.05). The optimal c_0 value was selected as the one that provided high overall classification accuracy and a balance between the

commission and omission errors. Pixels that were reclassified as “unmapped” based on the UDM2/UDM or labelled as “unmapped” in the interpreted 20 burned area maps were ignored. The classified 659 PlanetScope image pairs were examined visually for quality assessment purposes. This was undertaken because burned area classification errors, such as stripes at input-image granule boundaries and anomalous burning patterns (for examples, see Humber et al. (2019)), may remain undetected by the burned area accuracy assessment that was undertaken using a limited number of PlanetScope validation reference data sets (Section 3.4). Summary statistics of the classified burned area proportions were also derived for each of the 659 image pairs to check that unfeasible classified burned area proportions did not occur.

3.4. PlanetScope burned area accuracy assessment

The classified PlanetScope image pairs were compared with the 20 validation PlanetScope burned area reference data sets (Section 3.2.2). Recall that the 20 sets were selected across Africa and were different from those used to refine the U-Net. As before, the overall classification accuracy and the burn omission and commission errors were derived from a burned/unburned confusion matrix, ignoring unmapped 3 m

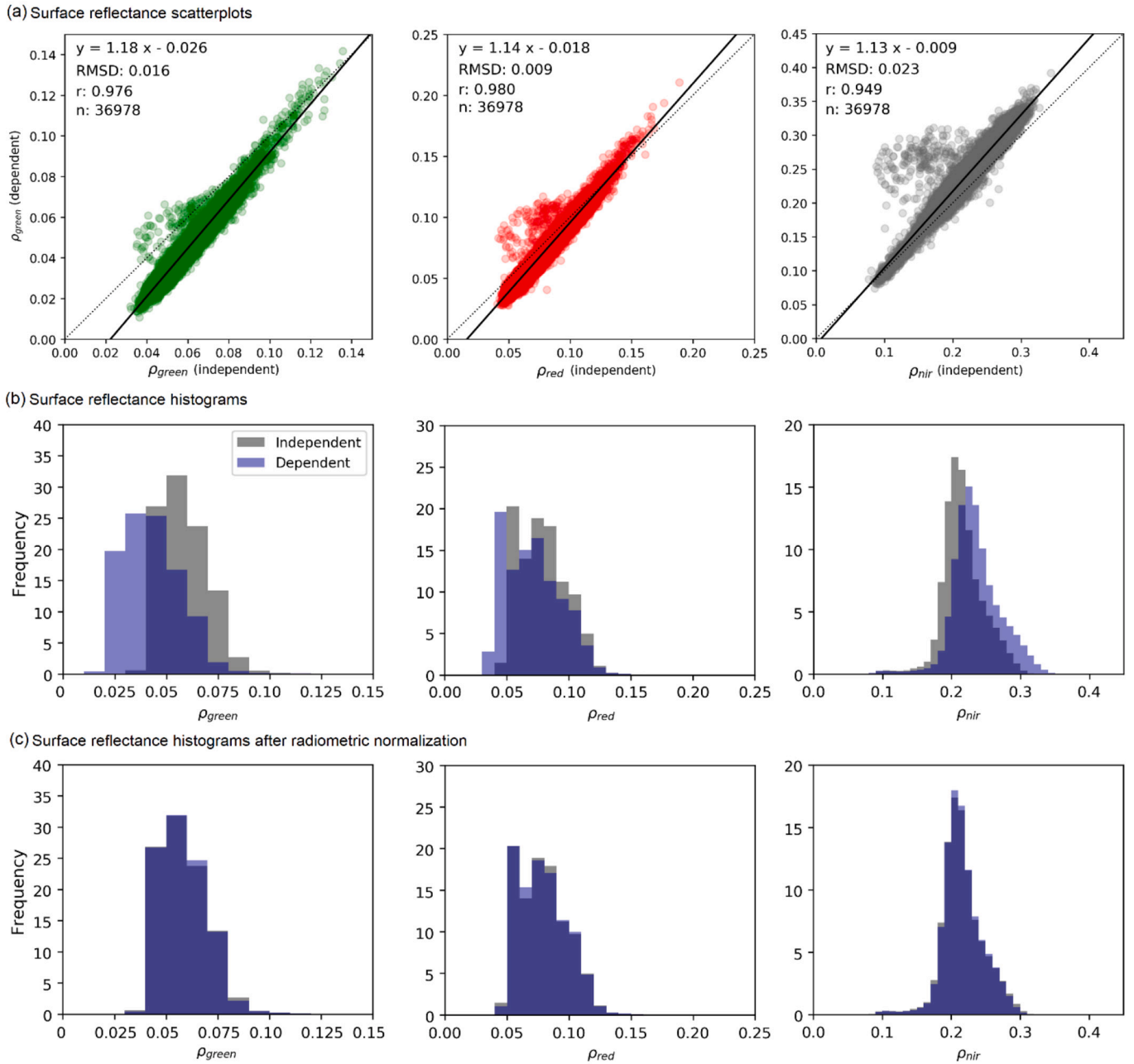


Fig. 7. (a) PlanetScope surface reflectance spectral scatterplots comparing the dependent image sensed June 22 2019 (Fig. 8a) and the independent image sensed June 23 2019 (Fig. 8b); surface reflectance spectral histograms before (b) and after (c) the relative radiometric normalization.

pixels.

The factor of ten scale discrepancy between the Landsat and PlanetScope imagery is potentially an issue for the transfer learning, as the spatial arrangement of burns in 30 m and 3 m pixel 256×256 patches may be different. Our expectation is that the substantial Landsat training data set used, and the U-Net refinement with PlanetScope imagery, may mitigate this issue. However, to examine this, the confusion matrix based accuracy measures were derived considering 256×256 3 m patches grouped by the proportion that was interpreted as burned. The patches were extracted from the 20 validation PlanetScope burned area reference data sets and categorized into eight proportion groups with the same number of patches per group. The same number of patches per group were extracted so that the accuracy results could be meaningfully compared among groups. For each group, the patches were extracted at random but ensuring that the patches did not overlap spatially, and that

each contained at least one reference mapped burns so that both omission and commission errors could be assessed. The number of patches that could be extracted that met these criteria was limited, and a total of 36 patches per group were selected.

In addition, the burned proportions defined in 30×30 m grid cells in the classified and the validation 3 m burned area maps were compared by linear regression, considering only grid cells with $<50\%$ unmapped data. The linear regression coefficient of determination (r^2) and the regression slope and intercept terms summarize the precision and the accuracy of the classified burned area proportions respectively (Roy and Boschetti, 2009; Boschetti et al., 2019). If the 3 m errors of omission and commission compensate each other in the 30 m grid cells, then the r^2 and slope term will be close to unity, and the intercept will be close to zero, indicating high precision and accuracy of the classified burned area proportions.

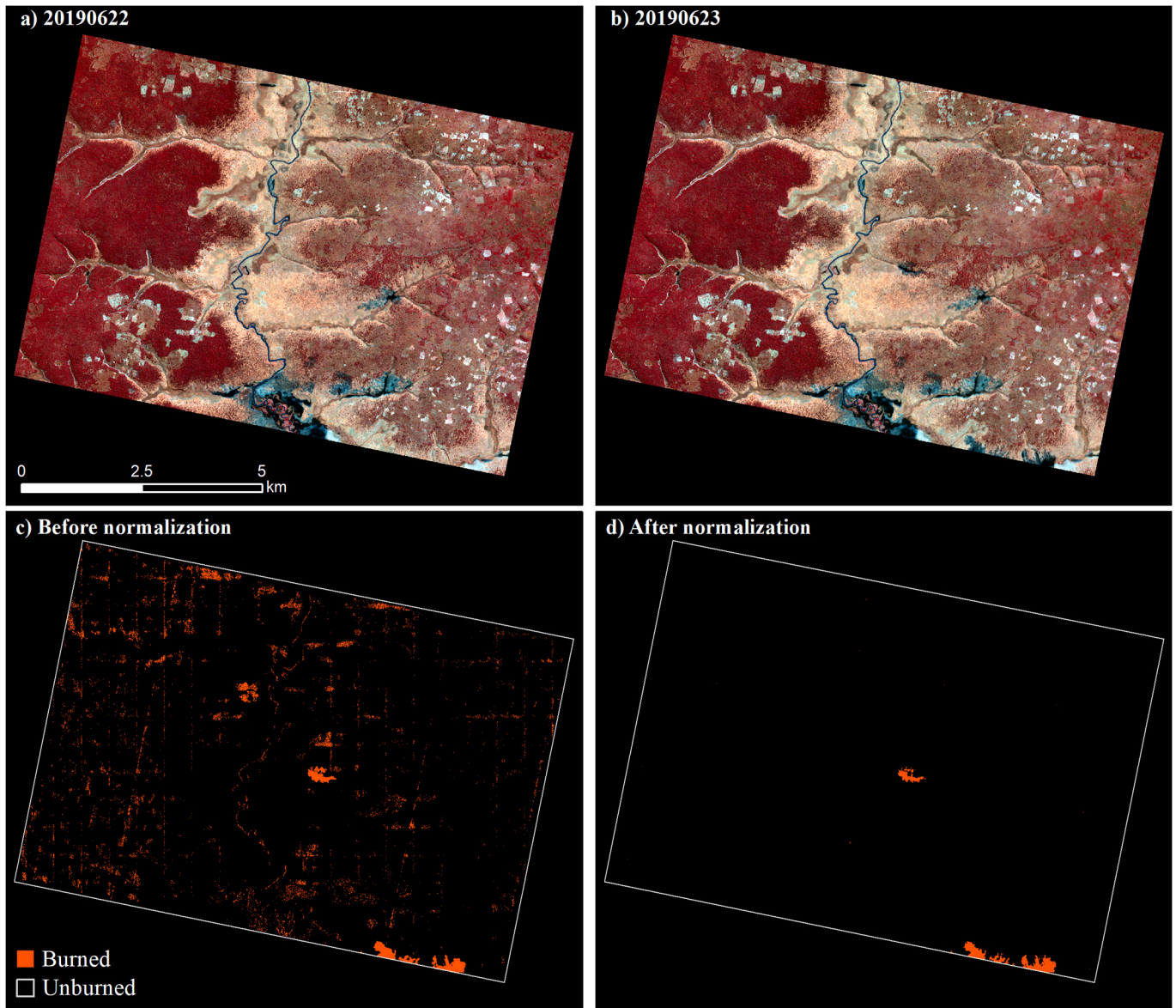


Fig. 8. Example PlanetScope 3 m image pair (NIR-Red-Green surface reflectance) sensed on (a) June 22nd 2019 (48.37° solar zenith, 0.22 AOD) and (b) June 23rd 2019 (48.4° solar zenith, 0.12 AOD), near Chitembo, Angola (13.37° S, 16.64° E). U-Net classification of the image pair (c) before and (d) after relative radiometric normalization. The U-Net was trained with 300,000 Landsat training patches. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4. Results

4.1. Landsat-8 burned area training and evaluation data

A total of 400,000 and 65,000 Landsat training and evaluation patches, respectively, were extracted from the 92 pre-existing Landsat-8 burned area reference data sets. Fig. 4 shows histograms of the patch burned area proportions. The training and evaluation histograms are similar, which indicates that the extraction methodology provided representative but independent sets of training and evaluation patches. The histograms are positively skewed (a greater frequency of small values); 90% of the patches have burned area proportions $<20\%$ and $<19\%$ and the most frequent proportions are 2% and 1%, in the training and evaluation datasets, respectively. $<1\%$ of the patches have burned area proportion $>50\%$ in either dataset.

4.2. Selection of optimal Landsat-8 burned area patch training set size and example U-Net Landsat burned/unburned classification results

Fig. 5 presents the results of the experiment designed to select an optimal amount of Landsat training data for the U-Net classification. An increasing proportion of the Landsat training patches was used to train the U-Net and classify the 65,000 Landsat evaluation patches. Eight proportions were considered, using from 5000 to all 400,000 of the training patches. The overall classification accuracies were all $>97.11\%$. The commission errors were relatively stable and varied by 3.3% among the eight sets of classifications but the omission errors decreased monotonically with the number of training patches from 22.47% (5000 training patches) to 13.15% (300,000 training patches) and then increased slightly to 13.86% (400,000 training patches). The small increases in omission error (0.64%) and commission error (0.54%) from 300,000 to 400,000 training patches indicates adequate training data representation provided by the 300,000 training patches. Further, the most balanced omission and commission errors were provided with

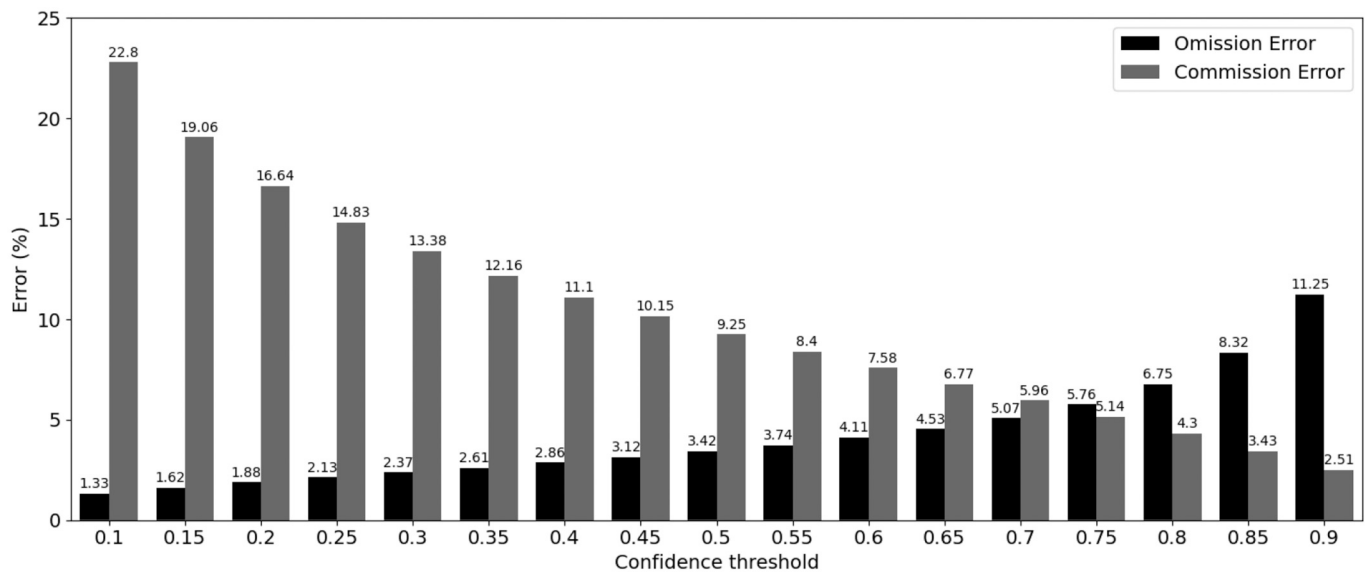


Fig. 9. Sensitivity analysis of the Refined U-Net classification accuracy with respect to the confidence threshold (c_0) used to classify the 20 PlanetScope burned area reference data sets used to refine the U-Net. The 20 PlanetScope image pairs were radiometrically normalized.

300,000 training patches. Consequently, the U-Net model trained with 300,000 training patches was selected for the rest of this study.

Fig. 6 shows examples of U-Net Landsat classification results for 128×128 30 m pixel subsets of three Landsat evaluation patches. Recall that the Landsat evaluation and training patches do not overlap (Fig. 2) and so the 300,000 training patches extracted across Africa to train the U-Net did not use the evaluation data. The relatively high degree of correspondence between the burned areas defined in the Landsat evaluation patches (Fig. 6c) and the U-Net classifications (Fig. 6d) is encouraging. Notably, the U-Net is able to handle variable pre- to post-fire reflectance changes, including quite subtle changes (Fig. 6 bottom row), and delineate complex burn boundaries. The Figs. 5 and 6 results indicate that the U-Net can be used to classify burned areas present in Landsat imagery and demonstrates the U-Net applicability on the source data (Landsat) before the PlanetScope transfer learning.

4.3. Illustration of the need for PlanetScope relative radiometric normalization

Fig. 7 illustrates the PlanetScope relative radiometric normalization process for an image pair, illustrated in the top row of Fig. 8, with a 75.5 km² common area acquired over central Angola. The first image (Fig. 8a) contains older burns in the center-south, with new burns along the S.E. edge and in the center of the second image that was sensed a day later (Fig. 8b). The Theil–Sen regression was implemented with the reflectance of the first image as the dependent variable and the reflectance of the second image as the independent variable because the first image had higher AOD (0.22) than the second (0.12). Fig. 7(a) shows scatterplots comparing the red, green and NIR surface reflectance for the two images and the Theil–Sen regressions (solid black lines). The new burns are exhibited by lower reflectance in the second (independent) image, particularly in the NIR, but, as the new burns occupied a minority of the overlapping image area they did not affect the Theil–Sen regression. The regression slopes and offsets are not unity and zero, respectively. This is likely because of residual atmospheric correction errors and sensor calibration differences between the two images, and not due to band-pass and bi-directional reflectance effects (Huang and Roy, 2021) as both were sensed by the same PlanetScope-1 sensor generation and with similar ($\sim 48^\circ$) solar zenith angles. Fig. 7(b) and 7(c) show surface reflectance histograms before and after the relative radiometric normalization, respectively, illustrating how the normalization makes

the reflectance for the dependent image more similar to the independent image.

The relative radiometric normalization improved the PlanetScope classification results or had no impact. For example, Fig. 8 shows the application of the U-Net trained with 300,000 Landsat patches, to the two PlanetScope images before (Fig. 8c) and after (Fig. 8d) normalization. In this example there is an evident reduction in the large number of commission burn errors (Fig. 8c) with only the new burns classified in the normalized results (Fig. 8d). The narrow intermittent lines of commission errors apparent in the unnormalized classification results (Fig. 8c) are likely due to the PlanetScope detector array boundaries (diagonal lines parallel to the image swath edges) and sensitivity to the burned area majority voting process in 256×256 3 m patch overlap areas (horizontal and vertical lines). These results also illustrate the utility of the transfer learning i.e., using a Landsat trained U-Net to classify the PlanetScope imagery, that is developed in the following sections.

4.4. Selection of the refined U-Net optimal burn confidence threshold and illustrative PlanetScope burned/unburned classification results

The refined U-Net was derived by further training the U-Net with 5000 3 m patches extracted from 20 PlanetScope burned area reference data sets (Section 3.3.3). Fig. 9 presents the results of the experiment to select the refined U-Net optimal burn confidence threshold value. This was undertaken by application of the refined U-Net to classify 20 pairs of PlanetScope images and burned area reference data considering a range of c_0 values (Section 3.3.4). Using a higher c_0 value will reduce the likelihood of a pixel being classified as burned. Thus, as expected, the omission and commission errors monotonically increase and decrease, respectively, with c_0 . However, this variation is not symmetrical around $c_0 = 0.5$. The overall classification accuracies were $> 99.4\%$ for all threshold values and the most balanced omission and commission errors, 5.76% and 5.14% respectively, were obtained for $c_0 = 0.75$. Consequently, 0.75 was selected as the optimal threshold value. These results were derived using the 20 PlanetScope burned area reference data sets used to refine the U-Net, and so they do not constitute an independent validation, which is presented in the following section.

Fig. 10 illustrates the improved classification results provided by the refined U-Net (Fig. 10d) compared to that provided by the U-Net (Fig. 10c). The top row of Fig. 10 shows a 5.6×6.1 km subset of a

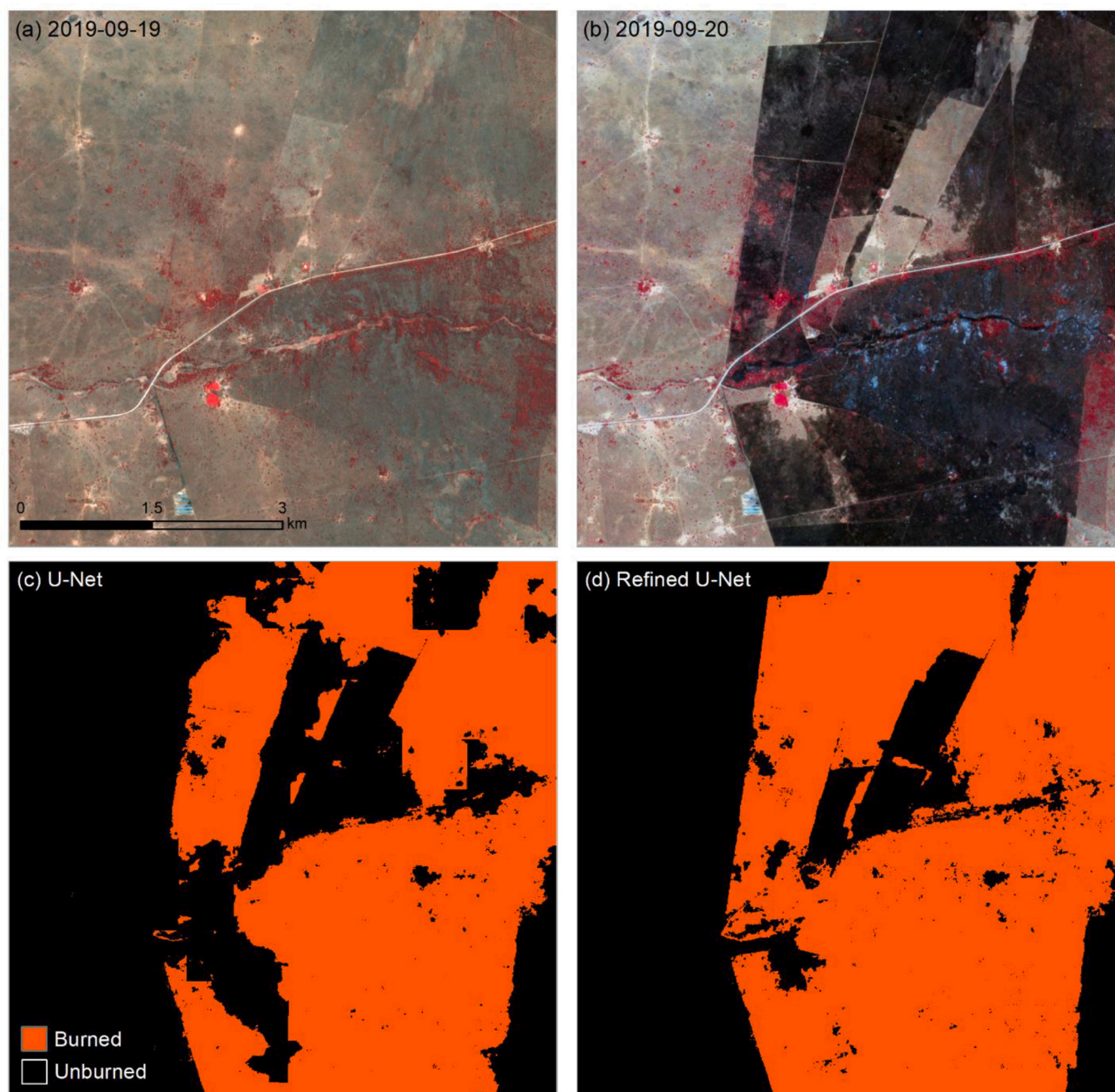


Fig. 10. Example radiometrically normalized PlanetScope 3 m image pair (NIR-Red-Green surface reflectance) sensed on (a) September 19th 2019 (40.8° solar zenith, 0.18 AOD) and (b) September 20th 2019 (37.9° solar zenith, 0.12 AOD), over a common area of 163 km², near Mahikeng, South Africa (25.84°S, 25.92°E), and the classification results derived (c) using the U-Net trained with Landsat training patches, (d) using the refined U-Net. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

PlanetScope image pair sensed a day apart falling in South Africa near the Botswanan border. This pair was selected because it captures an extensive biomass burning event that covers a large number of 3 m pixels (extensive black areas evident across the center and east of the second image, Fig. 10b) and because it includes burns with linear and curved edges (delimited by roads and field boundaries), burns with irregular edges, and faint burns that are indicative of low combustion completeness that are challenging to detect (Roy et al., 2019). Notably, comparing Fig. 10c and d, the refined U-Net was able to more precisely delineate the different burned boundaries including the boundaries of interior unburned areas. In addition, the refined U-Net was able to better classify the extensive faint burned areas occurring near the northern

image edge (just west of 25.94°E) and in the central south. These qualitative examples demonstrate the relevance of the fine-tuning procedure to provide improved transferability and burn classification accuracy.

4.5. Africa PlanetScope burned-unburned classification and validation results

The refined U-Net was used to classify the 659 pairs of radiometrically normalized PlanetScope images. The classified pairs contained a diversity of burns with individual burns ranging from single 3 m pixels to extensive burns that covered up to 16.6% of the common classified

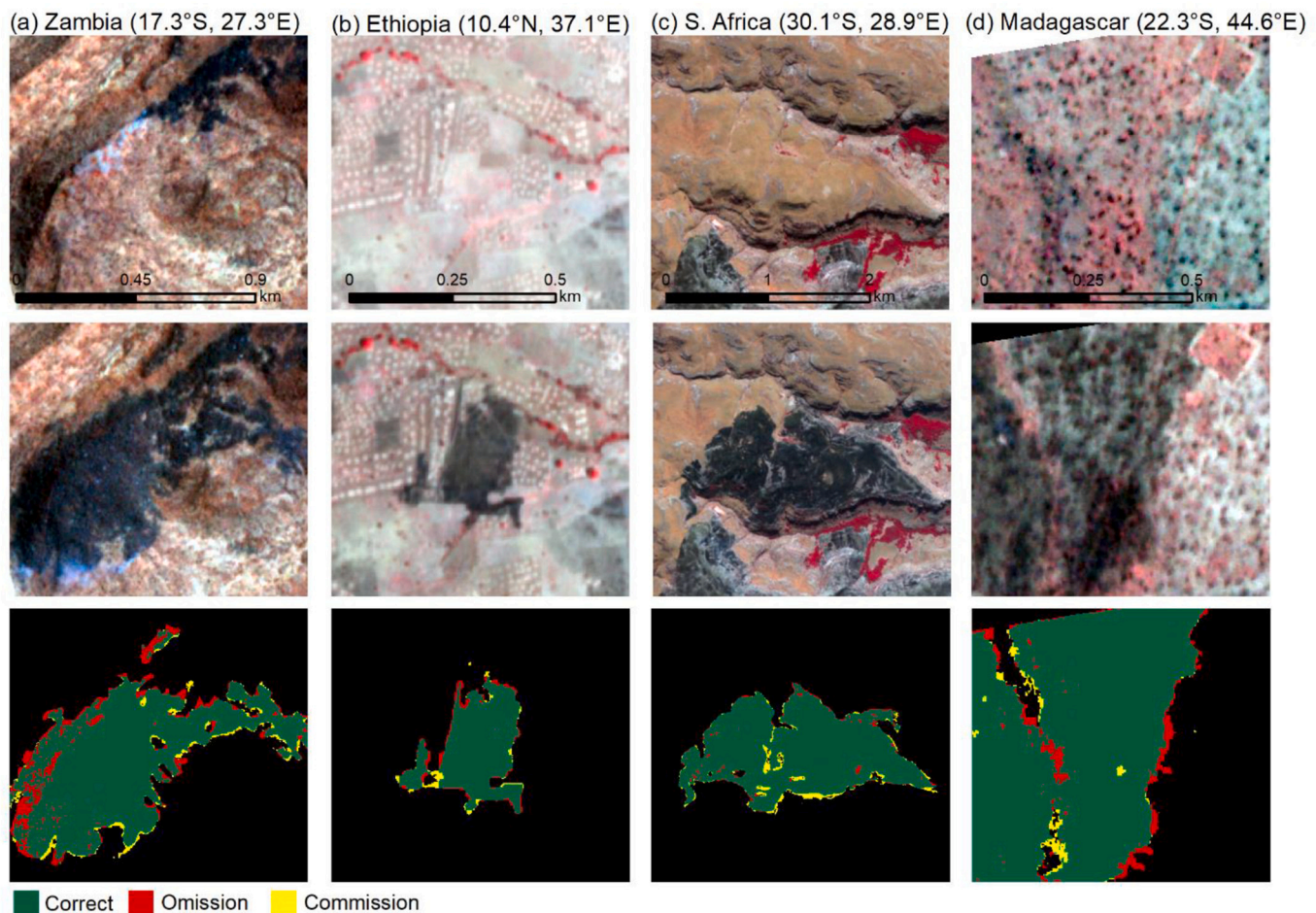


Fig. 11. Examples illustrating refined U-Net burned area classification results. The radiometrically normalized PlanetScope 3 m images (NIR-Red-Green surface reflectance) were sensed one day apart in 2019. The bottom row illustrates the 3 m classification and burned area reference maps derived for the validation together, showing correctly classified pixels (green), pixels that should have been classified as burned but were not (red), and pixels that were incorrectly classified as burned (yellow). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

image area, with a median of 0.09%. **Figs. 11 and 12** show illustrative classification results in different countries for eight of the 20 PlanetScope image pairs that were used to derive 3 m burned area reference maps for the validation ([Section 3.2.2](#)). **Fig. 11** shows examples with a high proportion of correctly classified pixels (green) relative to the burned area reference maps, and **Fig. 12** shows examples with larger proportions of omission (red) and commission (yellow) errors. Small spatial subsets (from 0.18 km² to 7.85 km²) with variable dimensions are illustrated to capture different burn patterns in detail.

In **Fig. 11**, the Zambian, Ethiopian and South African examples illustrate the capability of the refined U-Net to preserve inner unburned islands, and to correctly classify only the new burns evident in the second image. The Madagascar example is over an extensive low combustion completeness burn, that are often hard to detect, but was classified by the refined U-Net. These examples are quite representative with relatively small proportions of omission and commission errors that occur typically along the burn boundaries.

Fig. 12 presents detailed examples of omission and commission errors. It is well established that surface changes that are spectrally similar to burned areas (e.g. flooding, rapid vegetation senescence, shadows, snow melt) may be misclassified as burned ([Roy et al., 2005a](#)). The Morocco example has commission errors due to agricultural harvesting that is difficult to differentiate reliably from burning ([Hall et al., 2016](#); [Giglio et al., 2018](#)). The Chad example shows an area of riparian vegetation with high NIR and low visible surface reflectance (red tones) bisected by two narrow river channels that drain into a lake (cyan

tones). The commission errors are atypical of the 659 PlanetScope image pair classification results, as water bodies were normally correctly classified as unburned. However, in this case the water was sediment laden, indicated by monotonically increasing PlanetScope reflectance with wavelength and by high (~ 0.2 to >0.3) NIR reflectance, and the reflectance dropped in the second image in every band over the water. The commission errors occur in the narrow river channels and nearby pools, and not the lake, perhaps because of their small dimensions that were captured by the U-Net training. The Angola and Cote d'Ivoire examples show typical omission errors that occur because the change in reflectance between the two images was too small to be detected.

The validation results for the 20 pairs and their locations are shown in **Fig. 13**. The results were derived by per-pixel comparison of the refined U-Net classification and the 3 m burned area reference map ([Section 3.3.4](#)). The burn commission errors range from 1.1% to 76.1% (median 13.1%) and the omission errors range from 3.3% to 42.0% (median 13.8%). The worst results were at the Morocco site which had the smallest number of interpreted burned pixels among the 20 sites, with a high proportion of omission errors around the edges of spatially fragmented burns, with commission errors due to agricultural harvesting (example shown in **Fig. 12a**). Recall that image pairs with ≥ 25 km² common area and $\geq 75\%$ high-quality pixels (cloud- and shadow-free, no missing or suspect pixels) were used. The image pairs in Morocco, Cote d'Ivoire and Uganda had the smallest interpreted burned areas, 2129, 5082, and 4585 3 m pixels, respectively, corresponding to $<0.5\%$ of their common areas. The sites in South Africa and Nigeria had the

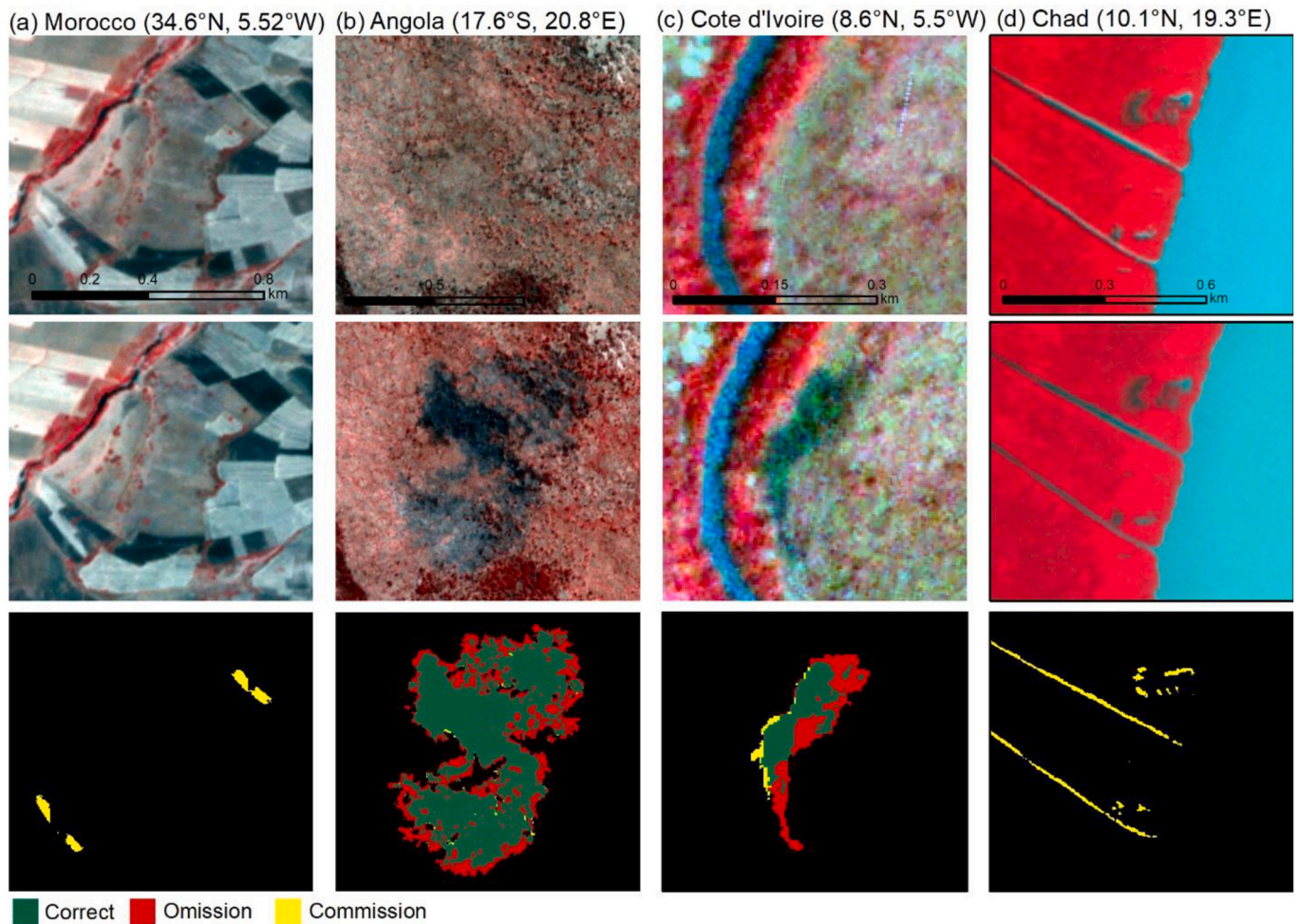


Fig. 12. Examples illustrating refined U-Net burned area classification results with errors. The radiometrically normalized PlanetScope 3 m images (NIR-Red-Green surface reflectance) were sensed one day apart in 2019. The bottom row illustrates the 3 m classification and burned area reference maps derived for the validation together, showing correctly classified pixels (green), pixels that should have been classified as burned but were not (red), and pixels that were incorrectly classified as burned (yellow). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

largest interpreted burned areas, 229,930 and 221,007 3 m pixels, respectively, corresponding to $<1.5\%$ of their common image areas. The overall classification accuracy for each image pair (not reported in Fig. 13) was always high ($\geq 99.65\%$) which is expected as only a minority of each reference map was burned.

Table 2 presents the confusion matrix generated considering all 20 site results together, composed of 339.4 million 3 m pixels. A total of 1522,97 pixels were classified as burned and 1,519,464 pixels were interpreted as burned in the reference maps. The overall classification accuracy was 99.89% which as noted above reflects the predominance of the unburned pixels and so is not a particularly useful accuracy measure (Stehman, 1997; Boschetti et al., 2004). The burn commission and omission errors were 12.29% and 12.09%, respectively, indicating a satisfactory per-pixel classification with balanced omission and commission errors.

Fig. 14 shows per-pixel omission and commission derived considering 256×256 3 m patches extracted randomly from the 20 validation pairs and grouped together by the patch proportion interpreted as burned. Eight proportion groups with 36 patches per group were extracted. The omission and commission errors decrease with the patch proportion interpreted as burned. The omission errors were 46% when the patches were > 0 to 0.1% burned and fell to about 10% when the patches were $> 6.5\%$ burned. The commission errors were 54% when the patches were > 0 to 0.1% burned and fell to about 6% when the patches were $> 15.8\%$ burned. These results indicate that the refined U-

Net classification of small burns is less reliable than larger ones. However, this finding is not definitive because patches with higher burned proportions can contain several disconnected small burns. For example, although the patch area is comparable to those illustrated in Figs. 12 and 13, patches may contain several different burns, for example, evident in Fig. 11a.

Fig. 15 shows a density plot of the proportion of 30 m grid cells (0–1) classified as burned against the proportion labelled as burned in the 3 m reference maps. The common areas of the 20 validation data sets were considered, >3.39 million 30 m grid cells. The rainbow logarithmic colour scale represents the frequency of cells with the same x-axis and y-axis proportion values. The values are predominantly clustered around the 1:1 line, reflecting the high degree of correspondence between the classified and reference data, and indicating that the errors of commission and omission compensate at 30 m resolution. Specifically, the low dispersion of the data, quantified by the coefficient of determination ($r^2 = 0.91$) is indicative of the precision of the classification, whereas the slope of the regression line close to unity (0.93) and intercept close to zero (< 0.001) reflect its accuracy (Roy and Boschetti, 2009; Boschetti et al., 2016). It should be noted that the relatively high density of cells along the edges of the plot, which indicates the presence of 30 m cells with large omission or commission errors, is primarily due to the omission and commission of small patches discussed above with respect to Fig. 14.

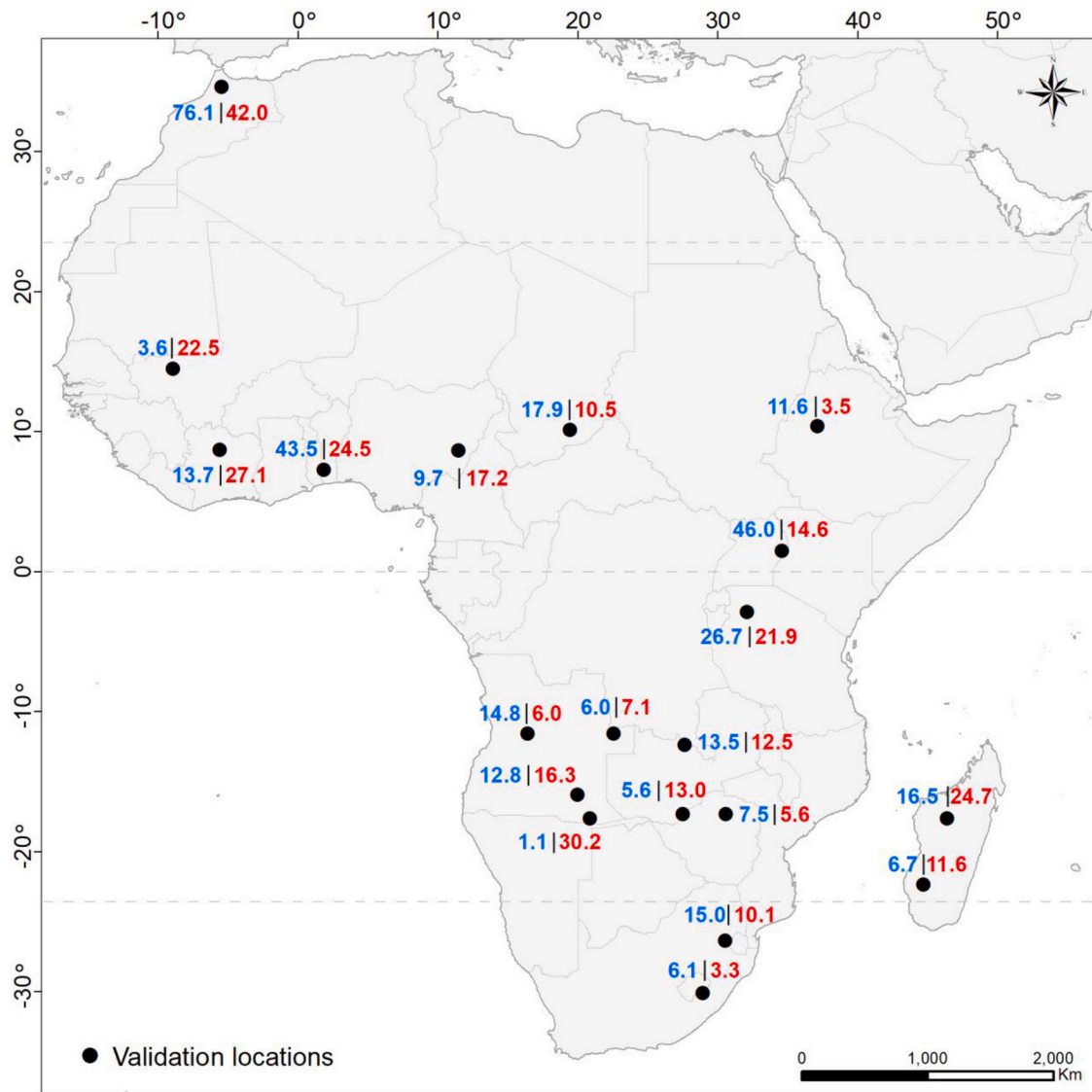


Fig. 13. Locations of the 20 3 m burned area reference maps used to validate the refined U-Net PlanetScope classification, and the resulting commission (blue) and omission (red) percentage errors. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2

Confusion matrix summarizing the refined U-Net classification accuracy derived from the 20 validation pairs (Fig. 13), matrix elements are shown as 3 m pixel counts and as percentages (in parentheses).

Classified	Reference		Total
	Burned	Unburned	
Burned	1,335,779 (0.39%)	187,198 (0.06%)	1522,977 (0.45%)
Unburned	183,685 (0.05%)	337,686,901 (99.50%)	337,870,586 (99.55%)
Total	1,519,464 (0.44%)	337,874,099 (99.56%)	339,393,563 (100.00%)

5. Discussion and conclusion

The near daily global coverage provided by the PlanetScope constellation opens new opportunities for high spatial resolution analyses (Roy et al., 2021), including burned area mapping. However, the sensors have no onboard calibration or SWIR bands, and are sensed with variable solar geometry, thus adding challenges to their use for large

area, automated, burned area mapping. To help overcome these issues, a U-Net deep learning algorithm was used to classify burned areas from two-date PlanetScope image pairs acquired at the same location. The U-Net architecture is trained with and applied to image patches and so incorporates spatial as well as spectral information. Given the significant effort required to develop a comprehensive burned area PlanetScope training data set, a transfer learning approach using pre-existing Landsat-8 derived burned area reference data (Boschetti et al., 2019) was instead developed.

Initial experiments using solely Landsat data confirmed that deep learning can be used to classify burned areas (Figs. 5 and 6). A conventional U-Net, trained with 300,000 Landsat-8 256 × 256 30 m patches, provided ~13% Landsat 30 m burn omission and commission errors that were quantified with 65,000 independent 256 × 256 30 m evaluation patches. As both the evaluation and classification data were 30 m resolution these results do not constitute a formal validation. The standard protocol for burned area validation emphasizes the importance of interpreted maps with finer spatial resolution than the classified product (Boschetti et al., 2009; Roy et al., 2019). Further work to investigate deep learning for Landsat burned area mapping is recommended and moderate resolution burned area mapping research, cited

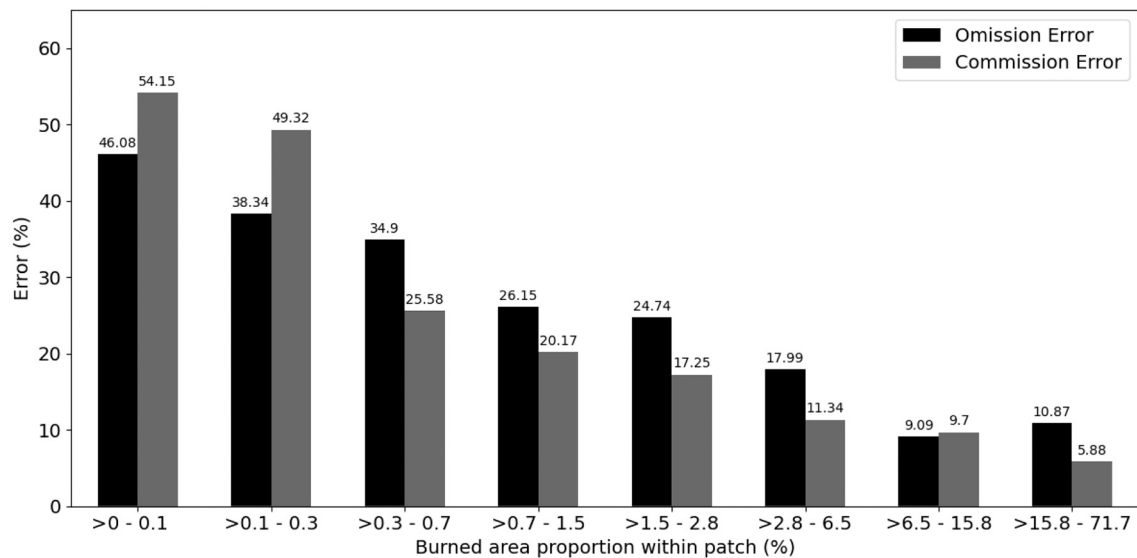


Fig. 14. Sensitivity analysis of the Refined U-Net classification accuracy with respect to burned area proportion. Results shown for eight proportion groups composed of $36,256 \times 256$ 3 m patches per group, that were randomly selected from the 20 validation pairs.

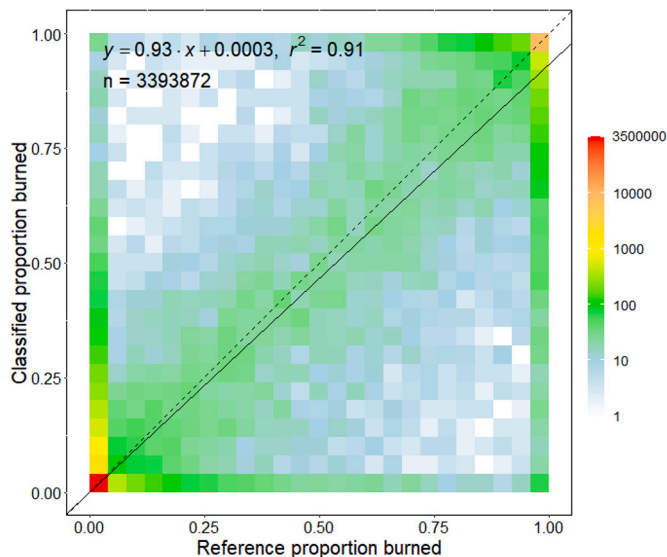


Fig. 15. Density plot of burned area proportions between the refined U-Net 3 m classified results and the 3 m reference data, within $30 \text{ m} \times 30 \text{ m}$ cells, considering the 20 validation data sets. The point density distribution is calculated using a 25×25 quantization of the plot axes, and is displayed with a rainbow logarithmic colour scale. The solid line shows the ordinary least squares regression of the plotted data. The dashed 1:1 line is shown for reference. Only grid cells with $<50\%$ unburned area were considered.

in the introduction, is ongoing.

A total of 659 PlanetScope 3 m image pairs sensed one day apart, with $\geq 25 \text{ km}^2$ common area, with $\geq 75\%$ high-quality pixels, and sensed with solar zenith difference $< 5^\circ$, were selected across Africa. It is well established that reliable burned area mapping requires stable reflectance time series, particularly when different sensor data are utilized (Roy et al., 2019). Atmospherically corrected and orthorectified (Level 3B) PlanetScope images were used but they do not have the same degree of temporal reflectance stability compared to more expensive earth observation systems such as Landsat (Houborg and McCabe, 2018). Consequently, each PlanetScope image pair was independently radiometrically normalized. A straightforward relative radiometric

normalization was applied based on a regression-based adjustment of the bands in the image with the higher aerosol optical depth (AOD) to the bands in the other image. This assumes that the lower AOD image has less aerosol contamination and so is more reliably atmospherically corrected (Ju et al., 2012; Doxani et al., 2018). Other normalization methods could be used. However, we found that the normalization unambiguously improved the transfer learning classification results (Fig. 8). Regardless of whether the data are radiometrically normalized or not, surface changes that are spectrally and spatially similar to burned areas may be misclassified as burned (Fig. 12). Future work to reduce burn commission errors is recommended, for example, based on time series deep learning, which is evolving rapidly (Masolele et al., 2021; Xu et al., 2018), with temporal constraints such as those used by the NASA MODIS and combined Landsat Sentinel-2 burned area product generation algorithms (Roy et al., 2019).

The implementation of a transfer learning approach presented several challenges. The Landsat-8 and PlanetScope imagery have different spatial (30 m vs. 3 m) and temporal (16 vs. 1 day) resolutions, different radiometric quality and spectral response functions (Houborg and McCabe, 2018; Huang and Roy, 2021), and the Landsat-8 and PlanetScope data were acquired in different years (2014–2015 vs. 2019). Despite these issues, the comprehensive training with 300,000 Landsat patches extracted across Africa likely captured a wide range of burned and unburned conditions, and the common features of burned areas, such as the drop in reflectance post-fire, likely helped a sensor agnostic U-Net burned/unburned representation. Previous transfer learning studies have highlighted the benefits of fine-tuning the trained network (Volpi and Tuia, 2016; Nogueira et al., 2017; Li et al., 2018). Consequently, in this study, further training of the Landsat derived U-Net model was undertaken using 20 PlanetScope burned area reference data sets. This involved additional training of the Landsat trained network layers with a small learning rate (Tremblay et al., 2018; Cetinic et al., 2018; Wurm et al., 2019). The resulting refined U-Net more precisely delineated burned boundaries and better detected faint burned areas (Fig. 10).

It is well established that statistically rigorous burned area product validation requires comparison with multi-date higher spatial resolution data that have been interpreted independently into burned area reference maps, and that have been collected with a suitable sampling strategy allowing for the estimation of accuracy metrics and relative standard errors (Padilla et al., 2015; Boschetti et al., 2016, 2019). However, there are no higher spatial resolution (i.e., $< 3 \text{ m}$) image pairs

acquired coincident with the PlanetScope image pair classification results reported in this study. Consequently, 20 PlanetScope image pairs, different to the 20 used to refine the U-Net, were visually interpreted to delineate 3 m burn reference maps. They were located across Africa and provide a Stage 2 validation (defined as in [Boschetti et al., 2016](#)) as they encompass a range of burned conditions but were not selected following a probability sampling design. In the future, the synoptic availability of very high spatial resolution commercial satellite data is anticipated that may provide a new source of burned area reference data, although significant effort is required to ensure the quality of the interpreted burned area reference maps.

The refined U-Net PlanetScope classification accuracy was first assessed using conventional 3 m per-pixel confusion matrix-based metrics. Balanced commission and omission errors of 12.29% and 12.09%, respectively, with variability among the 20 validation locations ([Fig. 13](#)) were observed. This overall performance is comparable or better than documented for Landsat and Sentinel-2 burned area mapping research ([Bastarrika et al., 2011](#); [Padilla et al., 2015](#); [Roteta et al., 2019](#); [Roy et al., 2019](#)) which as discussed earlier are sensors better suited for burned area mapping, although, meaningful comparison with other studies is difficult because of the different validation protocols, spatial coverage, and sensor resolutions that were employed. An experiment to examine the dependency of the U-Net classification accuracy on the burned area proportion within 3 m pixel 256×256 patches ([Fig. 14](#)) indicated that classification of patches with smaller burned area proportions was less reliable than larger proportions. This may be related to the scale difference between the Landsat training patches and the patches used to classify the PlanetScope imagery, although natural processes often exhibit spatial self-similarity over a range of scales ([Pentland, 1984](#)), and we often observed burned areas with similar spatial arrangements at Landsat 30 m and PlanetScope 3 m resolution. Object-based accuracy measures ([Persello and Bruzzone, 2010](#); [Yan and Roy, 2014](#)) were not used in this study as they are unreliable for objects comprised of only a small number of pixels ([Jaccard, 1901](#); [Tetteh et al., 2021](#)) and reliable clustering of burned pixels into burned objects is not trivial ([Roy et al., 2019](#)).

The per-pixel confusion-matrix results cannot discriminate between errors of omission and commission due to misclassification of entire burned or unburned patches, and errors due to the misclassification of isolated pixels in the vicinity of correctly classified burned patches. In-depth examination of these issues is beyond the scope of this paper. The regression analysis between proportion of area burned in coarser resolution cells showed however that the classification results, when aggregated at 30 m resolution, are both precise (as indicated by the high coefficient of determination) and accurate (as indicated by the regression line small intercept and slope close to unity). This implies that the omission and commission errors largely compensate at 30 m resolution, thus suggesting that the proposed classification methodology would be suitable for the generation of independent high resolution reference data for the validation of moderate spatial resolution continental and global burned area products. The results of this study indicate that systematic burned area mapping with PlanetScope imagery is possible, providing local, near daily, assessment of fire propagation, and mapping of small and spatially fragmented burned areas.

CRedit authorship contribution statement

V.S. Martins: Methodology, Software, Formal analysis, Visualization, Writing – original draft, Writing – review & editing. **D.P. Roy:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing. **H. Huang:** Visualization, Writing – original draft. **L. Boschetti:** Writing – original draft. **H.K. Zhang:** Writing – original draft, Writing – review & editing. **L. Yan:** Writing – original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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